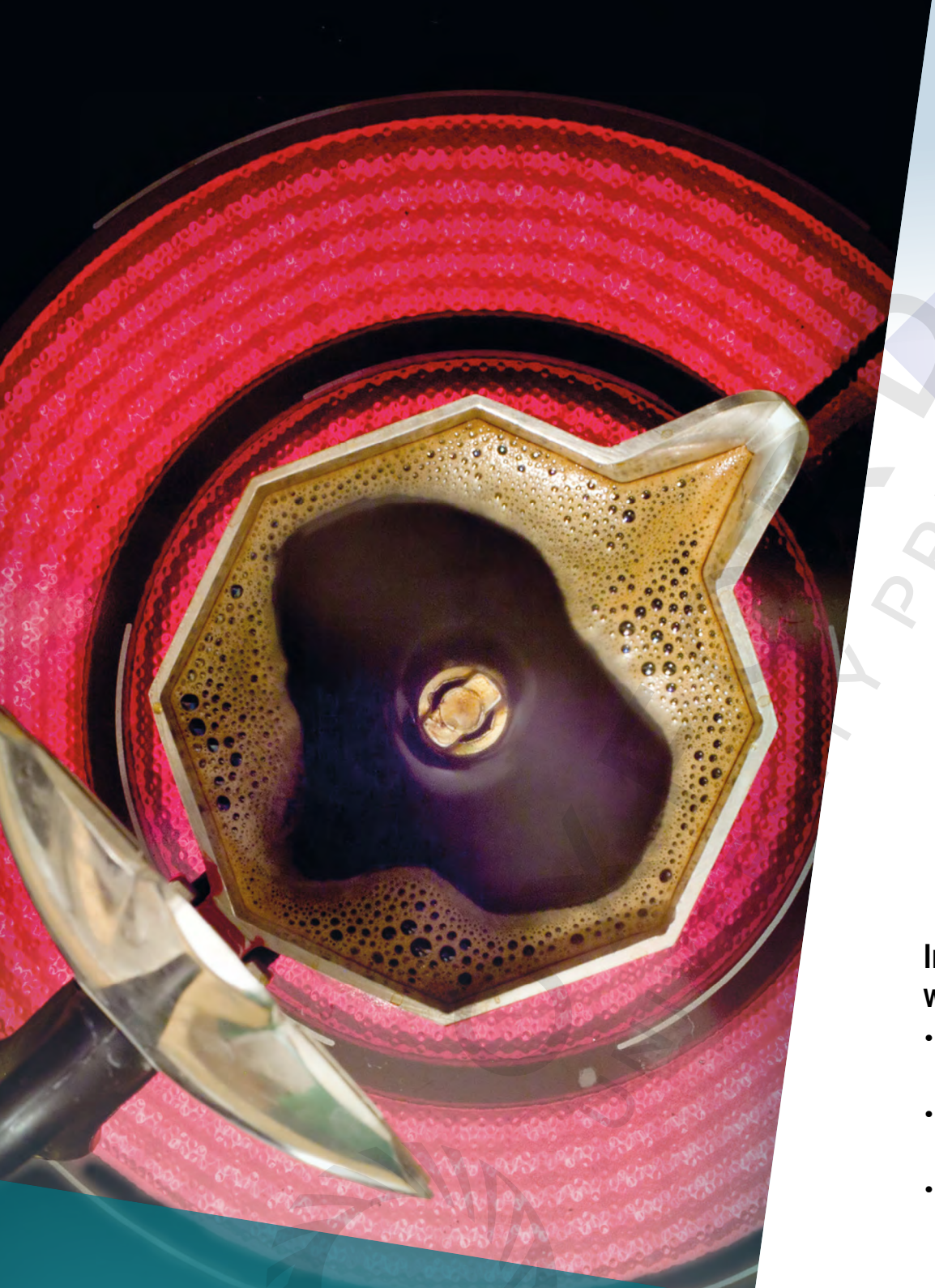


1

In this chapter, we will learn about

- the energy efficiency of electrical appliances
- the use of electricity in domestic cooking
- the working principle of air conditioners

Electricity at Home

1.1 Energy-consuming appliances

1 Main source of domestic energy

DSE goal

State electricity as the main source for domestic energy.

Flipped classroom

Watch a video and answer the questions.

What kinds of energy do the appliances in your home use (Fig 1.1a)?



Fig 1.1a *Some household appliances.*

You may find that most household appliances use electricity. Electricity is the main source of domestic energy. Let us look into the usage of electricity in our homes.

2 Energy efficiency

DSE goal

Describe the energy conversion involved in electrical appliances.

Define end-use energy efficiency in terms of the ratio of the amount of useful energy output to energy input.

DSE exam

15(2)Q3.4

When we use electrical appliances, different energy conversions occur (Fig 1.1b and 1.1c).

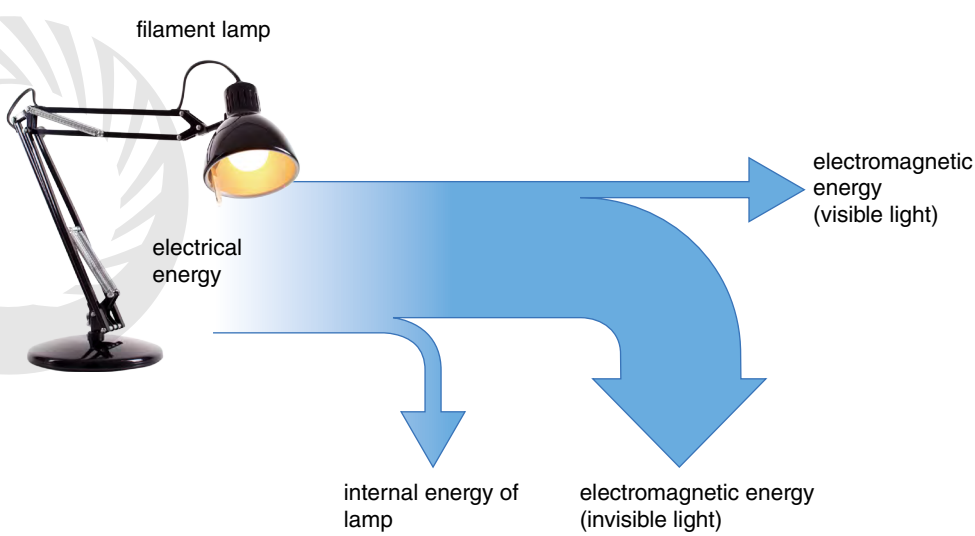


Fig 1.1b *Energy conversion in filament lamp.*

Electromagnetic energy is the energy of electromagnetic waves. A filament lamp emits visible light as well as other electromagnetic waves, such as infra-red radiation.

Remind Ss that energy conversions take place around us all the time: even while reading this book, our eyes are converting energy from light into electrical impulses which are transmitted along nerves to the brain.

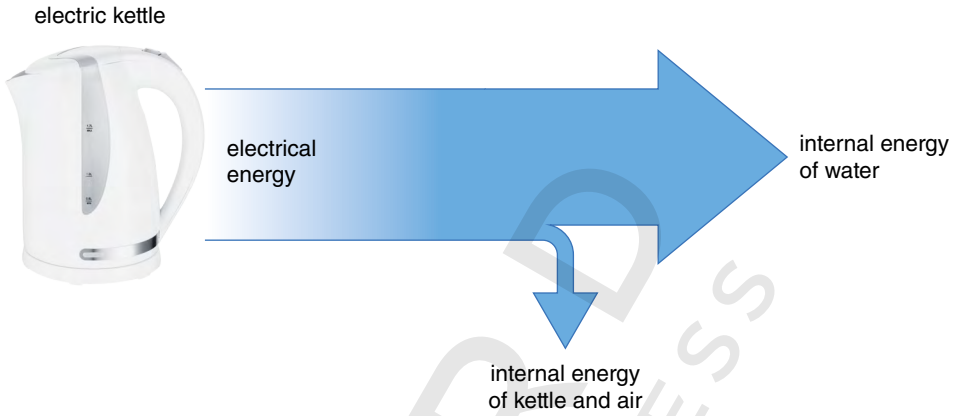


Fig 1.1c *Energy conversion in electric kettle.*

For example, when we use a filament lamp, electrical energy is converted into electromagnetic energy and internal energy. However, among these forms of energy only the energy of visible light is useful to us.

η is a Greek letter and is pronounced as 'eta'.

The **end-use energy efficiency** (η_{end}) of an appliance tells us what proportion of the energy input is converted into 'useful' energy by the appliance.

$$\text{End-use energy efficiency} = \frac{\text{useful energy output}}{\text{energy input}} \times 100\%$$

$$\eta_{end} = \frac{E_{out}}{E_{in}} \times 100\%$$

Caution

E_{out} represents the 'useful' energy output.

E_{in} represents the energy input to the appliance. Processes like the generation and transmission of electricity should not be considered.

For the same type of appliances, the one with a higher end-use energy efficiency uses less energy to do the same job (Fig 1.1d).



Fig 1.1d *Appliances with different end-use energy efficiencies.*

Tip

Since power is the energy transferred per unit time, the end-use energy efficiency can also be expressed as

$$\eta_{end} = \frac{P_{out}}{P_{in}} \times 100\%$$

end-use energy efficiency 最終能源效益

Know more

Internal energy

Do you know which of the following electrical appliances has the highest end-use energy efficiency?



electric kettle



electric space heater



electric lamp

The answer is the electric space heater. Do you know why?

Most electrical appliances convert some electrical energy into internal energy, which is usually not useful. But for an electric space heater, this becomes useful energy output. The heater essentially converts all the input electrical energy into the desired form, so its efficiency is higher than that of the other two appliances.

Example 1 | End-use energy efficiency of a kettle

A 2200-W electric kettle heats 1.5 kg of water with an initial temperature of 25 °C. The water starts to boil in 4 minutes.

(a) Find the energy input into the electric kettle.

(b) Find the useful energy output of the electric kettle.

(c) Find the end-use energy efficiency of the electric kettle.

Given: specific heat capacity of water = 4200 J kg^{-1} $°C^{-1}$

Solution

(a) Energy input $E_{in} = Pt$

$= 2200 \times (4 \times 60)$

$= 528\ 000$ J

Tip

The useful energy for an electric kettle is the energy absorbed by the water.

(b) Useful energy output $E_{out} = mc\Delta T$

$= 1.5 \times 4200 \times (100 - 25)$

$= 472\ 500$ J

(c) End-use energy efficiency $= \frac{E_{out}}{E_{in}} \times 100\%$

$= \frac{472\ 500}{528\ 000} \times 100\%$

$= 89.5\%$

Ask Ss how the efficiency of the electric kettle can be improved. (Improve the thermal insulation of the kettle to minimize energy loss; increase the power of the kettle to minimize the duration of energy loss.).

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.



1 What is the useful energy output of a sandwich maker?

A The energy consumed by the sandwich maker

B The energy released by the heating element of the sandwich maker

C The energy gained by the sandwich

2 A power station uses fossil fuel to generate electricity. Assume that 100 kJ of energy in fossil fuel is consumed and the resulted electrical energy is supplied to an electrical appliance at home. The useful energy output of the appliance is 20 kJ. Which of the following explain(s) why this 100 kJ cannot be taken as the energy input in calculating the end-use energy efficiency of the electrical appliance? Put a tick in the corresponding box(es).

☐ Energy is lost when electricity is generated in power station.

☐ Energy is lost when electricity is transmitted to the appliance.

☐ Energy is lost to the surroundings when the electrical appliance is operating.

Learn by practice 1

1 A cooker has an end-use energy efficiency of 93%. If its useful power output is 1600 W, what is its power input?

2 A 2000-W electric kettle has an end-use energy efficiency of 88%. If it is used to heat some water for 3 minutes, how much energy does the water gain?

DSE goal

Discuss the uses of the Hong Kong Energy Efficiency Labelling Scheme (EELS) for energy-saving.

Solve problems involving EELS.

DSE exam

13(2)Q3.7, 22(2)Q3.4

3 Energy labels

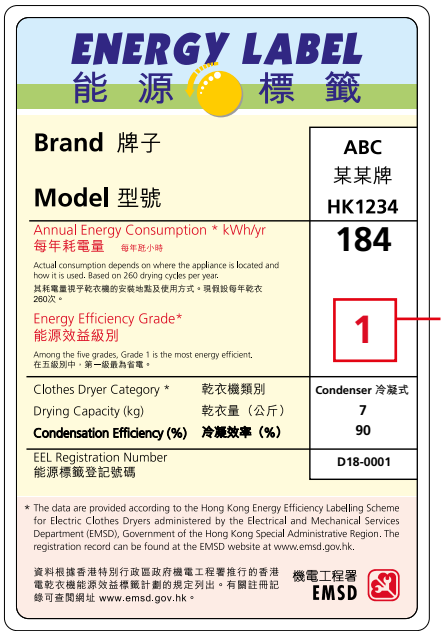
Electrical appliances with high energy efficiencies save energy. Electrical appliances included in the Hong Kong **Energy Efficiency Labelling Scheme**, or **EELS** for short, are affixed with energy labels to show their energy performance. There are two labelling schemes: voluntary and mandatory. The voluntary scheme includes 'Grading Type' and 'Recognition Type' energy labels, while the mandatory scheme includes only 'Grading Type' energy labels (Fig 1.1e on p.6).

The voluntary and the mandatory schemes each cover different types of appliances. You may visit the EMSD website for more details.

Useful link

Energy Efficiency Labelling Scheme (EELS) 能源效益標籤計劃

(i)

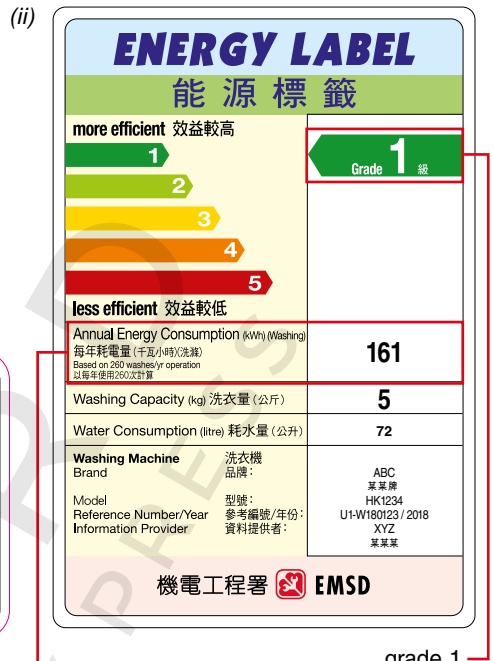


Grading Type label classifies the product's energy performance into grades 1 to 5. Grade 1 is the most efficient and 5 the least efficient. Grade 3 means the energy efficiency is about the average in that category.



Recognition Type label only indicates that the product has met the minimum energy efficiency specified in the scheme.

(ii)



161 kW h of energy is consumed if the washing machine is used for 260 washes per year

Fig 1.1e *(i) The 'Grading Type' and 'Recognition Type' energy labels in the voluntary scheme; (ii) the 'Grading Type' energy label in the mandatory scheme.*

Example 2 Energy label

A part of the energy label of a television is shown below.

Annual Energy Consumption (kWh) 每年耗電量 (千瓦小時) Based on 1460 hrs/yr operation 以每年使用1460小時計算	**38**
Diagonal Screen Size (cm) 屏幕對角尺寸(厘米)	**80**
Energy Efficiency Index 能源效益指數	**0.19**
Television 電視機 Brand 品牌 :	**ABC** 某某牌

(a) Estimate the cost of electricity in operating the television for 1460 hours. The cost of electricity is \$1.1 per kW h.

(b) Estimate the input power of the television.

Solution

(a) Cost of electricity = \$1.1 × 38 = \$41.8

(b) Input power = $\frac{E}{t} = \frac{38}{1460} = 0.0260 \text{ kW} = 26 \text{ W}$

Tip

According to the label, with the assumption that the television operates for 1460 hours in a year, it consumes 38 kW h of energy.

Quick check 2

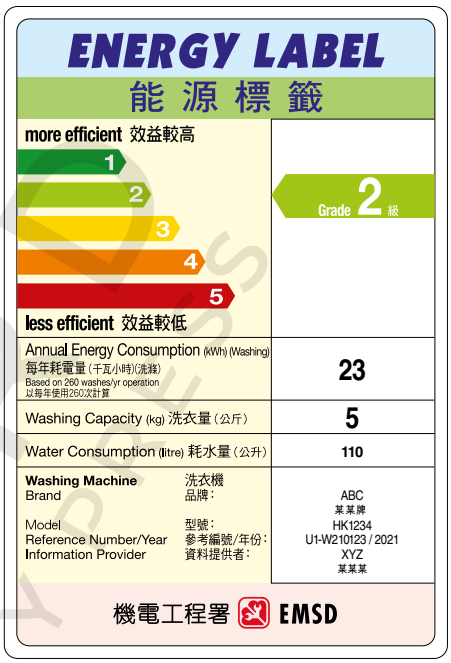
→ E-version in OUP Exercise Platform is also available.

1 Read the information from the energy label of a washing machine on the right and answer the following questions.

(a) What is the energy efficiency grade of the washing machine?

(b) Does the washing machine have an energy efficiency higher than, about the same as or lower than the average of other similar washing machines?

(c) If 260 washes are done in a year, how much energy is estimated to be consume in
(i) one year;
(ii) each wash?



Google Forms

Microsoft Forms

Exercise 1.1

1 The following energy label is affixed to a refrigerator (Fig a). What is the total amount of energy that the refrigerator consumes when it is used for 12 years?

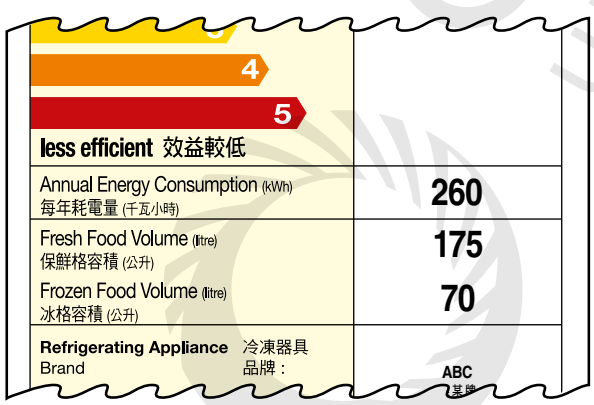


Fig a

A 260 J
B 260 kW h
C 3120 J
D 3120 kW h

★**2** Electric kettle *X* has a higher end-use energy efficiency than electric kettle *Y*. When the two kettles are used to heat the same amount of water from 25 °C to 100 °C, which of the following statements must be correct?

A *X* transfers more energy to the water.
B *X* has a higher output power.
C *X* consumes less energy.
D *X* finishes heating in less time.

★**3** Two electric rice cookers *X* and *Y* have end-use energy efficiencies of 70% and 80% respectively. If the cookers have the same useful energy output of 500 kJ, the energy input of *X* will be greater than that of *Y* by

A 50 kJ.
B 89 kJ.
C 400 kJ.
D 5000 kJ.

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*4 A 300-W conveyor can transport 600 kg of goods from the ground level to a platform 1.2 m above the ground in 1 minute (Fig b).

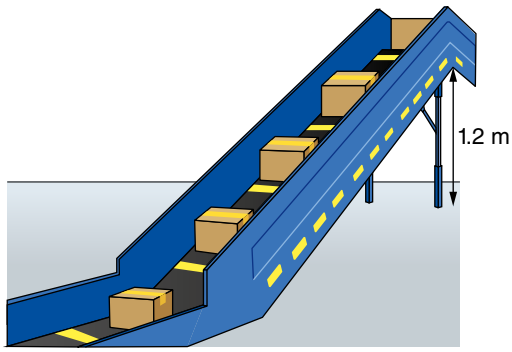


Fig b

(a) Suggest three types of energy involved when the conveyor operates.

(b) Calculate the end-use energy efficiency of the conveyor.

*5 Two electric kettles, *A* and *B*, have the same power input. The end-use energy efficiencies of *A* and B are 85% and 76% respectively. To boil 1 kg of water at 25 °C, if *A* takes 10 minutes, how long would *B* take?

*6 A toy boat is powered by a battery (Fig c). When it is turned on, its propeller rotates and it accelerates on the surface of the water.



Fig c

(a) What form(s) of energy is the kinetic energy of the propeller converted to?

(b) The e.m.f. of the battery is 3 V and the current drawn from it is 0.1 A. The mass of the boat is 0.4 kg. It accelerates from rest to a speed of 0.8 m s^{-1} in 2 s. What is the end-use energy efficiency of the boat?

*7 A 2-kW electric kettle is used to heat 500 g of water at 25 °C. It takes 12 minutes to boil away all the water. Take the specific heat capacity of water as 4200 J kg^{-1} $°C^{-1}$ and the specific latent heat of vaporization of water as 2.26×10^6 J kg^{-1}.

(a) Find the useful energy output, i.e. the energy absorbed by the water.

(b) Hence estimate the end-use energy efficiency of the kettle.

(c) Indeed, the heating element of the kettle is not completely immersed in water in the last minute of heating. How does this affect the efficiency of the kettle? Explain your answer.

Operating an electric kettle without water inside could be very dangerous. Do not try it.

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1.2 Cooking without fire

Electric cookers are becoming more popular nowadays. Figure 1.2a shows some of them. Do you know how they generate heat to cook?



Fig 1.2a Some common electric cookers.

DSE goal

Describe how electric hotplates, induction cookers and microwave ovens work in heat generation.

DSE exam

18(2)Q3.4

1 Common types of electric cookers

In the following part, we will introduce three common types of electric cookers: **electric hotplates**, **induction cookers** and **microwave ovens**, and their principles in generating heat.

a Electric hotplates

An electric hotplate makes use of the heating effect of current directly. It has a coil of resistance wire that acts as a heating element (Fig 1.2b). Electrical energy is converted into internal energy of the resistance wire at a rate of $P = I^2R$, with I being the current passing through it and R being the resistance of the wire.

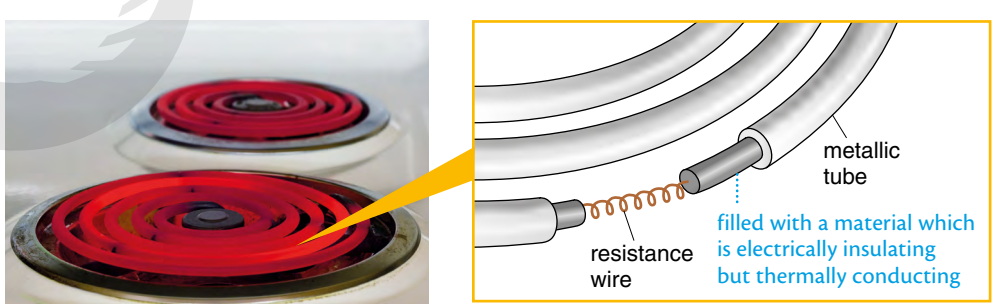


Fig 1.2b Electric hotplates with resistance wire encased in a metallic tube.

electric hotplate 電熱平板爐　　induction cooker 電磁爐　　microwave oven 微波爐

Figure 1.2c shows the working principle of an electric hotplate.

① A current passes through the resistance wire and produces a strong heating effect.

② Heat is transferred to the pot by conduction or radiation.

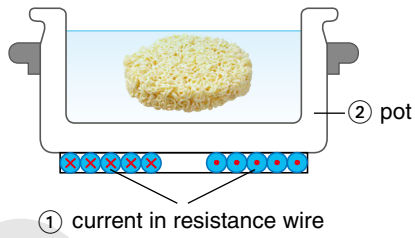


Fig 1.2c *Working principle of an electric hotplate.*

The metallic tube in an electric hotplate is shaped into a flattened coil and may be covered with a metal top plate (Fig 1.2d). This provides a large surface area for the transfer of heat to the cookware by conduction. Some newer models of electric hotplates have flat and smooth top plates made of glass-ceramic (Fig 1.2e). In this type of hotplate, heat is transferred from the heating element to the cookware mainly by radiation.

Glass ceramic is a poor heat conductor but it allows infra-red radiation to pass through effectively.



Fig 1.2d *An electric hotplate with a metallic top plate.*



Fig 1.2e *An electric hotplate with a glass ceramic top plate.*

All cooking pots with flat bases are suitable for electric hotplates. This is because the flat base have good contact with the heating element. This allows efficient heat transfer.

Video

→ This video shows how an induction cooker works.

b Induction cookers

An induction cooker (Fig 1.2f) induces a current in the metal base of a cooking pot to produce a heating effect. It contains a large coil of wire which is placed under a non-metallic top plate (Fig 1.2g).



Fig 1.2f *An induction cooker.*



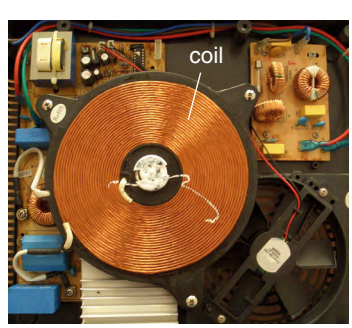


Fig 1.2g *The coil inside an induction cooker.*

As the induced e.m.f. and the subsequent induced current are directly proportional to the rate of change of the magnetic field, an induction cooker must use an a.c. with a high frequency.

Cross-link

Refer to Book 4 Ch 7.2 for the formation of eddy currents.

Figure 1.2h shows the working principle of an induction cooker.

① A high-frequency alternating current (a.c.) passes through the coil and sets up a fast-changing magnetic field.

② The metal base of the cooking pot experiences the changes in magnetic field continuously.

③ Due to electromagnetic induction, eddy currents are induced in the metal base to oppose the change in the magnetic field.

④ The eddy currents produce a strong heating effect in the metal base. Heat is then transferred to the food inside the pot by conduction.

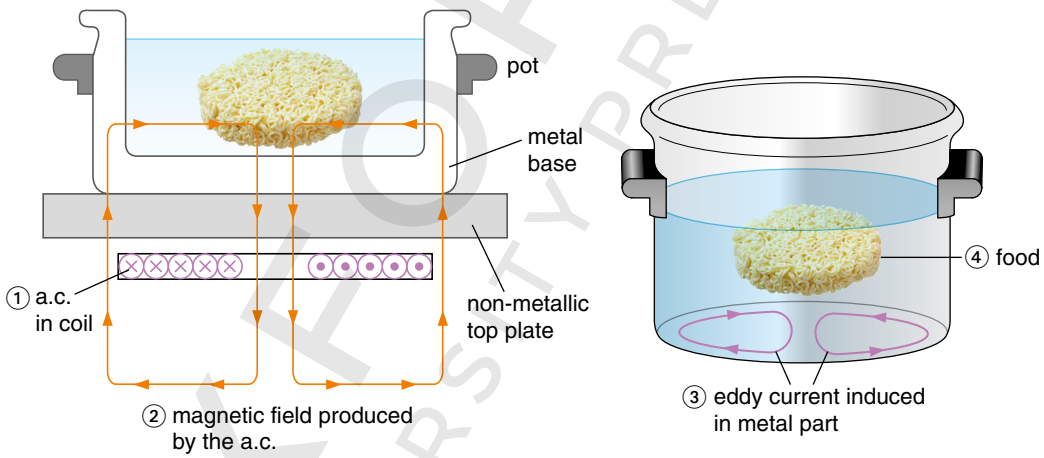


Fig 1.2h *Working principle of an induction cooker.*

Theoretically, eddy currents can be induced in non-ferromagnetic metals (such as copper and aluminium). However, since most induction cookers are designed to detect ferromagnetic pots only, metal pots which are not ferromagnetic cannot be used.

Cooking pots with metallic bases allow closed current paths to form and produce the heating effect for cooking. In general, pots made of ferromagnetic materials (such as iron and some kinds of stainless steel) work the best since they strengthen the induction effect (Fig 1.2i).

The top plate of the induction cooker is an insulator. Eddy current cannot form in the top plate so it does not get hot itself (Fig 1.2j). However, the top plate still gets hot owing to heat transfer from the pot by conduction.

Watch a video to see that the top plate remains cool when an induction cooker is cooking.

Video



Fig 1.2i *Cooking food with an iron pan on the induction cooker.*

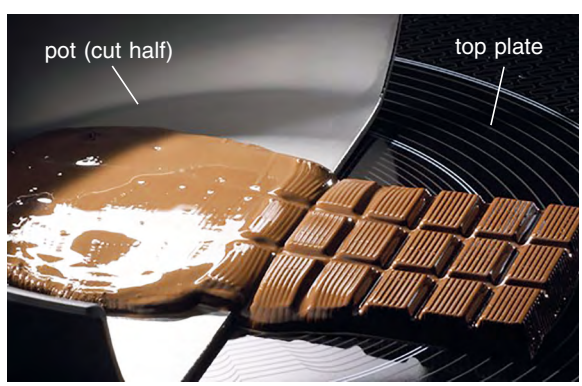


Fig 1.2j *The top plate of an induction cooker does not get hot.*

C Microwave ovens

The major components of a microwave oven include a cooking chamber, a *magnetron tube* and a *waveguide* (Fig 1.2k). Figure 1.2l shows the working principle of a microwave oven.

① Microwaves (usually at 2.45 GHz) are produced by rapidly oscillating electrons in the magnetron tube.

② The microwaves are led to the cooking chamber by the waveguide (a metallic tube).

③ The microwaves penetrate the food to cook them.



Fig 1.2k *A microwave oven.*

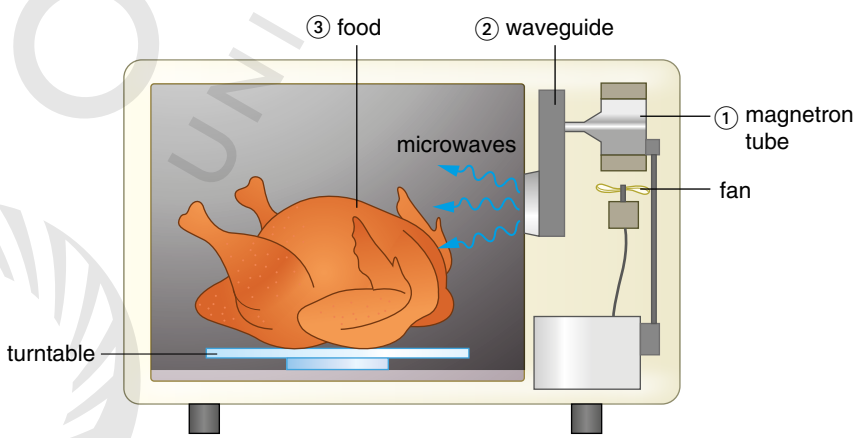


Fig 1.2l *Working principle of a microwave oven.*

Know more

Percy Spencer (1894–1970)

Percy Spencer invented the microwave oven in 1945. He thought of the idea after a candy bar melted in his pocket while he was working on a radar set emitting microwaves.

magnetron tube 磁控管　　waveguide 波導管　　Percy Spencer 史賓沙

Other polar molecules, such as sugars and fats, can absorb microwaves in the same way as water molecules do. Many substances containing polar molecules can be heated up by microwaves.

✎ Food with low water content may be difficult to be heated up by a microwave oven. On the other hand, food with high water content can be heated up quickly (in terms of power), but the rate of increase in temperature may still be low due to the high specific heat capacity of water.

Food can be heated up by microwaves effectively if it contains water. This is because water molecules are *polar*, i.e. they are positively charged at one end and negatively charged at the other (Fig 1.2m).

The polar water molecules flip rapidly as they keep aligning themselves with the oscillating electric field of the microwaves (Fig 1.2n). As they flip, they collide with other nearby molecules and make these molecules vibrate faster. The average kinetic energy of the food molecules and hence the temperature of the food increase.

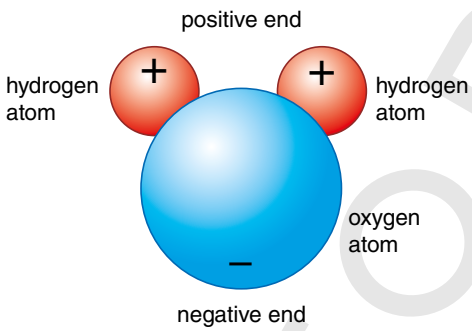


Fig 1.2m *A water molecule.*

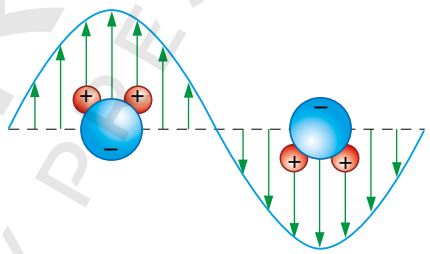


Fig 1.2n *The oscillating electric field makes the water molecules flip rapidly.*

Since microwaves of frequency 2.45 GHz can penetrate deeply into food (generally a few cm), the interior and the surface of the food can be heated up at the same time. Therefore, the food is heated up more quickly compared with a conventional oven (Fig 1.2o).

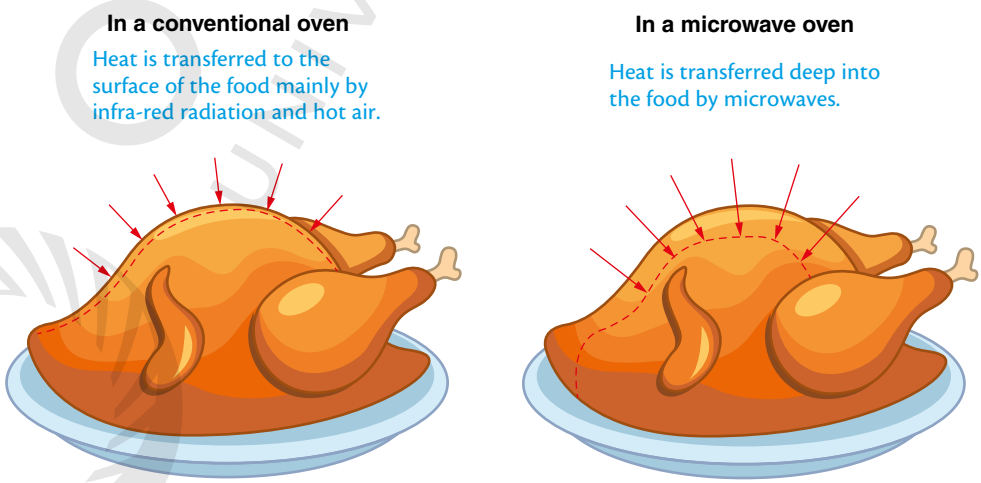


Fig 1.2o *Food heated in a conventional oven and a microwave oven.*

A microwave oven has a metal grid on the door. It blocks microwaves but allows visible light to pass through (Fig 1.2p on p.14). As the metal grid and the metal walls reflect microwaves, microwaves are confined to the chamber for cooking food.

polar 帶極性

A wave cannot pass through a gap that is much shorter than its wavelength. The diameter of a hole in the metal grid ($\sim 10^{-3}$ m) is much shorter than the wavelength of the microwave ($\sim 10^{-1}$ m) but much longer than the wavelength of visible light ($\sim 10^{-7}$ m).

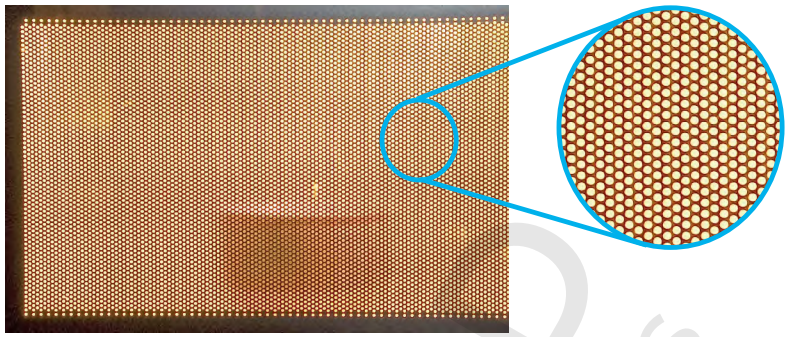


Fig 1.2p *Metal grid on the door allows visible light but not microwaves to pass through.*

Cross-link

Refer to Book 3B Ch 5.4 for details of stationary waves.

When microwaves are reflected, they meet other trains of microwaves, and stationary waves may be formed (Fig 1.2q). The part of food at the nodes of the stationary waves cannot be heated effectively. To solve this problem, a microwave oven usually contains a turntable that keeps rotating the food during heating so that no part of the food will stay at a node throughout the heating process.

The locations of the nodes are the cold spots in the microwave oven, while the locations of the antinodes are the hot spots.

Watch a video to see the structure and working principle of a microwave oven, and also a demonstration of the hot spots and cold spots in a microwave oven using cheese.

Video

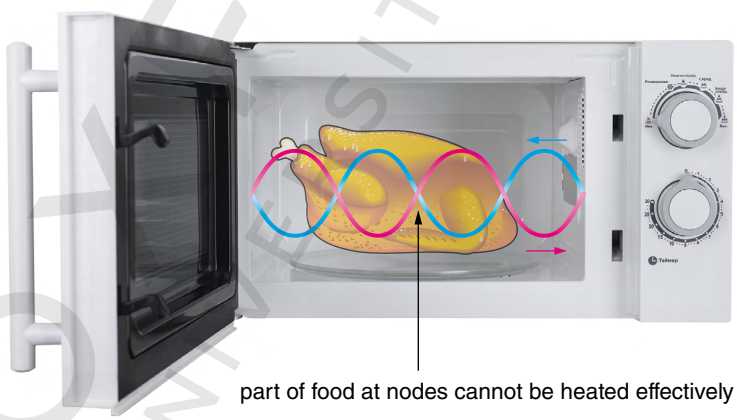


Fig 1.2q *Microwaves inside the cooking chamberforming stationary waves.*

Sharp metal objects (e.g. metal fork) are dangerous to use in microwave ovens. Under the electric field of microwaves, electrons are concentrated at the sharp ends and sparks may be produced.

Know more

Using microwave ovens safely

The following lists some safety precautions in using microwave ovens.

- Do not put metals or other conducting materials into the oven. These materials may produce sparks that can damage the oven and cause a fire.
- A microwave oven may cause a liquid to superheat, i.e. reaching a temperature higher than its boiling point without boiling. The liquid can boil explosively when you take the container out of the oven. Superheating can be avoided by putting a microwave-safe spoon into the liquid when heating it.
- Do not heat whole eggs or food in closed containers in microwave ovens. These items can explode due to the pressure built up by steam generated inside the enclosed space.

OUP eResources

The following summarizes how different types of electric cookers generate heat.

Electric hotplate

- **makes use of the heating effect of current in a coil of resistance wire to generate heat**

Induction cooker

- **uses an a.c. to set up a changing magnetic field, which induces eddy currents in the metallic cooking pot. Heat is then generated due to the heating effect of the eddy currents**

Microwave oven

- **converts electrical energy into microwaves. Polar water molecules in the food flip rapidly due to the oscillating electric field of the microwaves. They collide with nearby molecules, causing the total KE of the molecules in the food to increase and so the food is heated up**

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Where is heat mainly generated when food is cooked by each of the following electric cookers?

		The electric cooker itself	The base of the cooking pot	Inside the food
(a)	**Electric hotplate**			
(b)	**Induction cooker**			
(c)	**Microwave oven**			

2 Which of the following is/are the main process(es) involved when food is heated by an induction cooker? Put a tick in the corresponding box(es).

- ☐ Heating effect of current
- ☐ Electromagnetic induction
- ☐ Production of microwaves
- ☐ Condensation of steam

3 Why can food containing water be heated up effectively by a microwave oven?

A Water has a high specific heat capacity.

B Water has a high specific latent heat of vaporization.

C Water molecules are polar.

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DSE goal

Solve problems involving end-use energy efficiency of cookers.

Discuss the advantages and disadvantages of electric hotplates, induction cookers and microwave ovens.

Use the power rating of cookers to determine running cost.

DSE exam

17(2)Q3.3, 20(2)Q3.4

2 Choosing electric cookers

The three types of cookers mentioned above generate heat through different processes. Figure 1.2r shows the energy conversions for each type of cooker and compares their end-use energy efficiencies.

Unlike an electric hotplate, the coil of wire in an induction cooker does not get very hot during operation, so not much energy is lost from it. This makes an induction cooker more energy efficient than an electric hotplate.

The magnetron tube in a microwave oven converts only about 70% of the input energy into microwaves. This makes the oven less energy efficient.

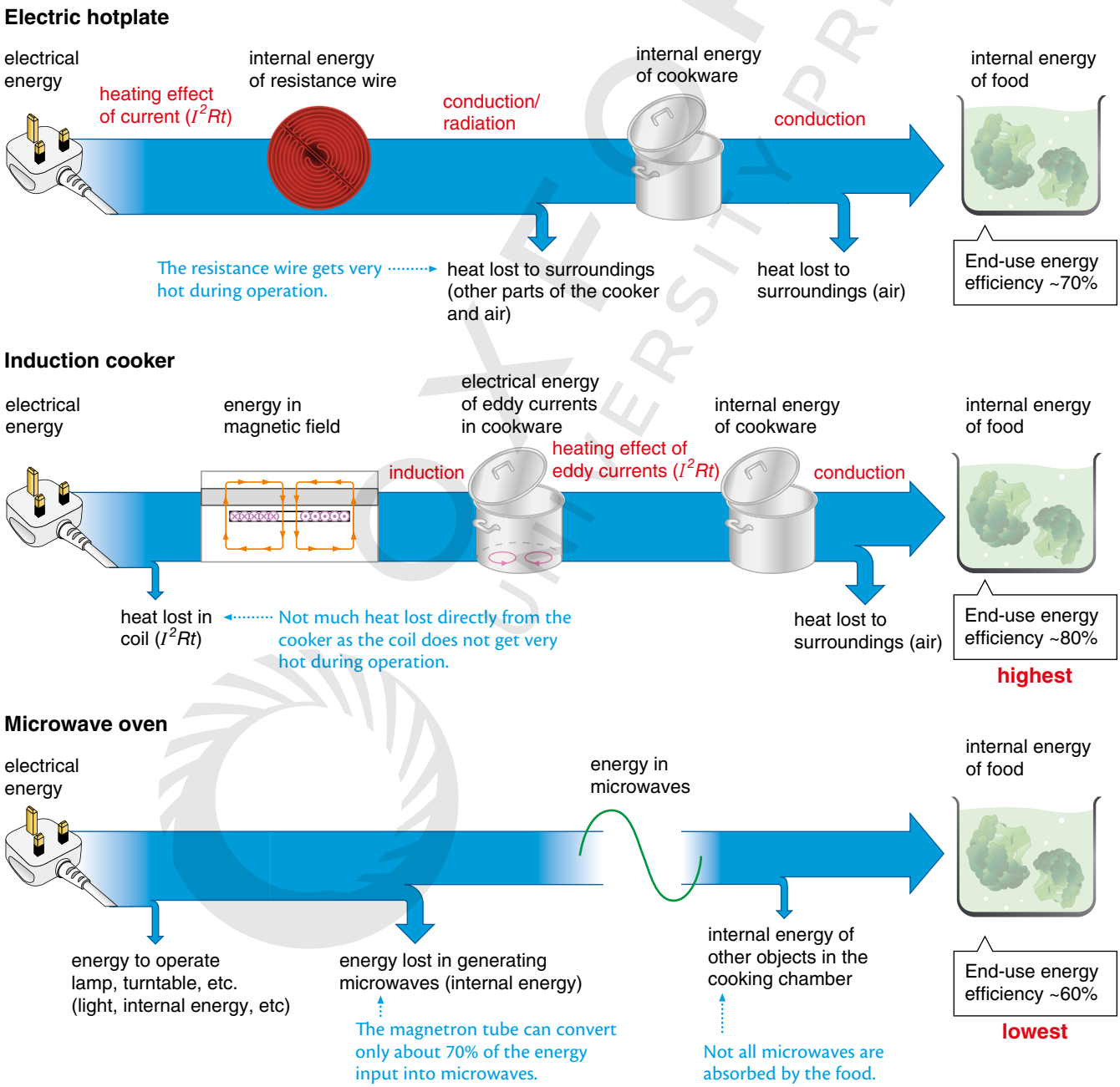


Fig 1.2r *Energy conversions for electric hotplate, induction cooker and microwave oven.*

✎ The energy saved in cooking methods is negligible when compared with the energy consumption of air conditioning and lighting.



These efficiencies only tell us what percentage of the energy input is available to use for heating. The actual amount of energy used in cooking depends on the cooking method (frying, boiling, steaming, etc.), the type of food and the cookware used. Also, note that the efficiency for every model of cooker differs depending on the design and materials.

Example 3 Energy output of a microwave oven

(a) A 800-W microwave oven takes time t to heat 0.3 kg of potato from 4 °C to 50 °C. The end-use energy efficiency of the oven is 55% in this process. Find t. The specific heat capacity of the potato is 3400 J kg^{-1} °C^{-1}.

(b) Paul suggests using an induction cooker since it has a high end-use energy efficiency. He immerses the potato in a pot of water and puts the pot on the induction cooker operating at 800 W. However, he finds that the actual time taken to heat the potato is much longer. Why?

Solution

(a) Energy needed $E = mc\Delta T = 0.3 \times 3400 \times (50 - 4) = 46\ 920$ J

Useful power output $P_{out} = P_{in} \times \eta_{end} = 800 \times 55\% = 440$ W

$$t = \frac{E}{P_{out}} = \frac{46\ 920}{440} = 107 \text{ s}$$

(b) Extra energy is required to heat the pot and the water. The amount of extra energy is large because water has a high specific heat capacity.

Tip

By $E = Pt$, $t = \frac{E}{P}$

P here is the rate of energy transferred to the potato and is the useful power output.

Example 4 Running cost

Daisy finds that the end-use energy efficiency of an electric hotplate is 72%. On average, 5.4 MJ of energy is needed for cooking each day. Calculate the cost of electricity per day for using the electric hotplate. The cost of electricity is $1.1 per kW h.

Solution

Energy input $E_{in} = \frac{E_{out}}{\eta_{end}} = \frac{5.4 \times 10^6}{72\%} = 7.5 \times 10^6$ J

Cost of electricity $= \$1.1 \times \frac{7.5 \times 10^6}{1000 \times 3600} = \$2.29$

Tip

Remember to convert J to kW h.

1 kW h = 1000 W × 3600 s
= (1000 × 3600) J

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Table 1.2a shows a comparison of the three types of electric cookers. It helps us choose a suitable type of cooker that best fits our requirements.

Electric hotplate	Induction cooker	Microwave oven
Working principle		
heating effect of current in coil → conduction / radiation to pot → conduction to food	heating effect of eddy currents induced in the metal base of cooking pot → conduction to food	flipping of water molecules in food due to microwaves, producing heating effect → conduction within food
Advantages		
• most pots with flat bases are suitable to use • low selling price	• high energy efficiency • top plate does not get very hot	• ideal for reheating (Fig 1.2s) • its case does not get very hot
Disadvantages		
top plate gets very hot	pots made of ferromagnetic materials are usually required	low energy efficiency

Table 1.2a *Comparing different types of electric cookers.*



Fig 1.2s *Microwave ovens are ideal for reheating cooked food.*

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Which forms of energy are involved when an electric hotplate is used to cook food? Put a tick in the corresponding box(es).

- ☐ Electrical energy of eddy currents
- ☐ Internal energy of coil in cooker
- ☐ Internal energy of cookware
- ☐ Internal energy of air

2 Which of the following is an advantage of an induction cooker over an electric hotplate?

A It has no special requirement on the cookware.
B It has a flat top plate.
C The top plate does not get very hot.
D It is good for reheating.

Exercise 1.2

1 Which of the following types of pots works best on an induction cooker?

A Ceramic pot
B Glass pot
C Copper pot
D Iron pot

2 Which of the following statements about an induction cooker is correct?

A Electric current flows from the coil in the cooker to the pot.
B The electric current flowing through the coil in the cooker generates heat.
C It is operated by a steady d.c.
D Eddy currents are induced in the top plate.

3 Which of the following statements about an electric hotplate is correct?

A It must be used with a pot made with a ferromagnetic material.
B It is safe to touch its top plate when in use.
C It operates by the heating effect of electric currents.
D No energy is lost to the surroundings when it is in use.

***4** Which of the following will occur when microwave passes through a water molecule (Fig a)?

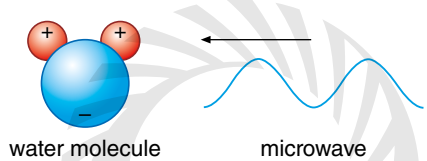


Fig a

(1) The molecule moves along the travelling direction of the microwave.
(2) The molecule flips repeatedly.
(3) The molecule gains energy.

A (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

***5** Which of the following statements about a microwave oven is/are correct?

(1) Metal walls and metal grids on the door are used to reflect and confine microwave inside the oven.
(2) It heats up the interior of the food before its surface.
(3) Metal containers should not be put into a microwave oven.

A (1) only
B (3) only
C (1) and (2) only
D (1) and (3) only

***6** Vegetables are cooked using an electric hotplate (Fig b) and a microwave oven (Fig c).



Fig b



Fig c

(a) How are the cooking pan and the container shown suitable to use for the hotplate and the microwave oven respectively?
(b) Suggest an advantage of each cooking method over the other one.
(c) State one safety precaution for each cooking method.

***7** A 1500-W electric hotplate and a 2000-W induction cooker have end-use energy efficiencies of 70% and 80% respectively. The cost of electricity is $1.1 per kW h. The average energy needed for cooking is 3 MJ each day.

(a) Find the difference in the daily cost of using the cookers.
(b) The induction cooker is $500 more expensive than the electric hotplate. How long will it take for the difference in the cost of electricity to compensate for the difference in the price between the cookers?
(c) Give one reason why an induction cooker has a higher end-use energy efficiency.

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1.3 Moving heat around

The air released by an air conditioner to the outside is very hot (Fig 1.3a). Where does the internal energy of the air come from? In this part, we will discuss how heat is moved around by an air conditioner.

Unless otherwise specified, air conditioners are used for cooling in this book. ▶

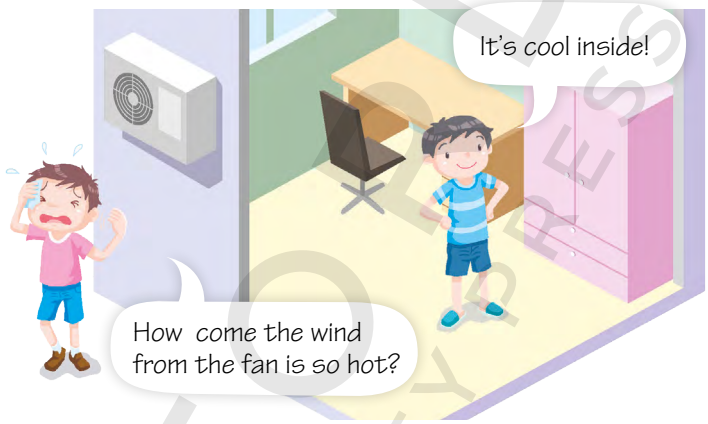


Fig 1.3a Hot air is released from an air conditioner.

DSE goal

Describe how air conditioner as a heat pump transfers heat against its natural direction of flow.

DSE exam

15(2)Q.3(c), 17(2)Q.3(a), 20(2)Q3.7

1 How an air conditioner works

A glass of hot water becomes cool after it is left on a table for some time. The reverse **never happens naturally**—we never see hot water becoming even hotter if we just leave it at room temperature.

The natural direction of heat flow is from a high-temperature body to a low-temperature one. If we want to transfer heat from a cold body to a hot body, we need to use a **heat pump** (Fig 1.3b).

A heat pump is a device that transfers heat against its natural direction of flow.

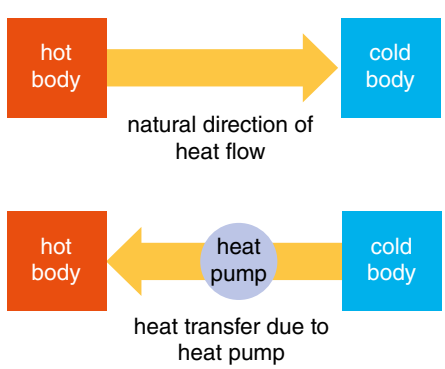


Fig 1.3b Heat transferred by heat pump against natural direction of heat flow.

heat pump 抽熱機

The air conditioner is a typical example of a heat pump. Figure 1.3c shows the basic structure of a typical window-type air conditioner.

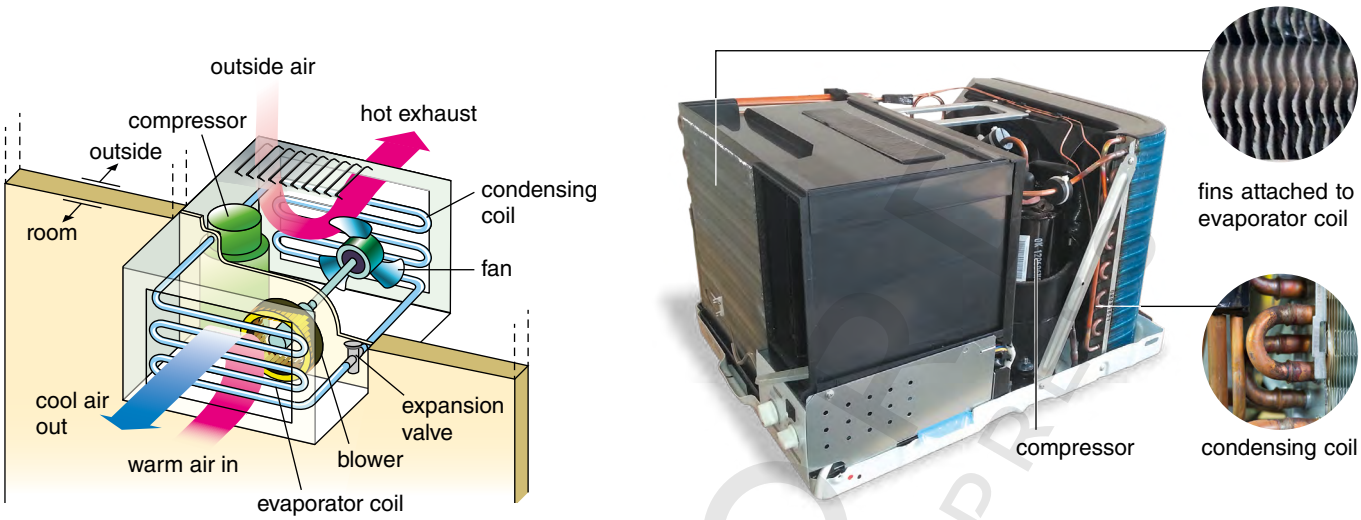


Fig 1.3c *The basic structure of a typical window-type air conditioner.*

Chlorofluorocarbon (CFC) was once commonly used as a refrigerant. However, this compound depletes the ozone layer, which shields the earth from the strong ultraviolet radiation from the sun. Nowadays, CFC is being replaced by other environmental friendly compounds, such as HFC.

The major components of an air conditioner include the *compressor*, the *condensing coil*, the *expansion valve* and the *evaporator coil*. They are connected by tubes to form a closed loop. A coolant called a *refrigerant* is driven by the compressor to flow in the loop.

The evaporator coil and the condensing coil are places where heat transfer occurs. They are highly coiled metal tubes with many metal fins attached to them. This design increases the contact area with the surrounding air to facilitate heat transfer. Fans and blowers are also used to assist the heat exchange processes.

Figure 1.3d shows the processes that occur as the refrigerant passes through the loop in one cycle.

Simulation

→ This simulation shows how an air conditioner works. It is available on OUP web in PC version (.exe) only.

The evaporator coil is inside the room while the condensing coil is outside the room. ▶

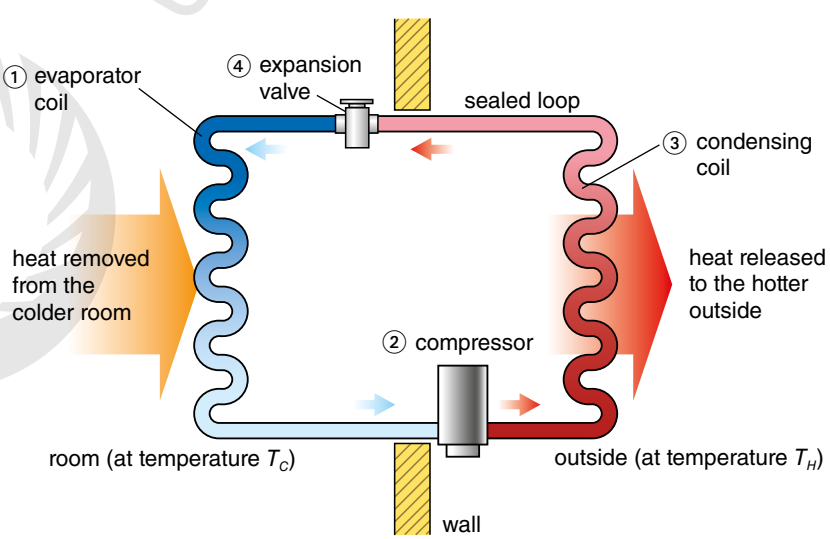


Fig 1.3d *Cyclic process in an air conditioner.*

compressor 壓縮機　condensing coil 冷凝管道　expansion valve 膨脹閥
evaporator coil 蒸發管道　refrigerant 製冷劑 / 雪種

① Evaporator coil

The refrigerant in the form of a liquid-gas mixture is colder than the air inside the room. The liquid refrigerant **absorbs latent heat of vaporization from the air in the room** and **evaporates**. This lowers the temperature in the room.

② Compressor

This is like when a bicycle tyre is pumped up, the gas pressure and gas temperature increases. ▶

The compressor compresses (does work on) the gaseous refrigerant. This **increases the pressure and the temperature of the refrigerant**. The refrigerant becomes hotter than the air outside the room.

✎ When the compressor does work on the refrigerant vapour, a force is required to move through a distance against the pressure of the vapour.

✎ Gaseous refrigerant at a higher pressure is more likely to condense.

③ Condensing coil

The hot gaseous refrigerant **releases latent heat of vaporization to the air outside the room** and **condenses to liquid**.

Cross-link

The latent heat of vaporization is mainly drawn from the liquid refrigerant itself, so the refrigerant is cooled. You learned the cooling effect of evaporation in Book 1 Ch 4.2. ▶

④ Expansion valve

The liquid refrigerant is at **low pressure** after passing the expansion valve. Part of the refrigerant evaporates, making the refrigerant colder than the air inside the room. The refrigerant then enters the evaporator coil and stage ① repeats.

Table 1.3a summarizes the above cyclic process in an air conditioner.

	Stage ①	Stage ②	Stage ③	Stage ④
Component passed	Evaporator coil 	Compressor 	Condensing coil 	Expansion valve 
State of refrigerant	liquid-gas mixture → gas	gas	gas → liquid	liquid → liquid-gas mixture
Temperature of refrigerant	increases, but colder than the room	increases, becomes hotter than the outside	decreases, but hotter than the outside	decreases, becomes colder than the room
Function	absorb heat from the room	increase the temperature of refrigerant ($> T_H$)	release heat to the outside	reduce the temperature of refrigerant ($< T_C$)
Internal energy of refrigerant	increases molecular PE ↑ molecular KE ↑	increases (due to work done by the compressor) molecular PE ↓ molecular KE ↑	decreases molecular PE ↓ molecular KE ↓	approximately unchanged molecular PE ↑ molecular KE ↓

Table 1.3a *Different stages in a cycle.*

✎ Increase the pressure of refrigerant to facilitate condensation

✎ Reduce the pressure of refrigerant to facilitate evaporation

By the above cyclic process, the air conditioner acts as a heat pump to transfer heat against its natural direction of flow.

During the cyclic process, heat Q_C is removed from the colder room, and heat Q_H is released to the hotter outside. The air conditioner has done work W to make this happen. From the law of conservation of energy (Fig 1.3e),

$$Q_H = Q_C + W$$

In other words, the amount of heat released to the outside is greater than the heat removed from the room, i.e. $Q_H > Q_C$.

Ts may tell Ss that the energy conversions of an air conditioner are as shown in the figure below. However, to solve the questions in this book, we usually need to take the air conditioner's electrical power input and work done to be equal. Ss may get confused that why the two quantities are equal even there is energy loss. Ts may tell Ss that this is just an assumption for calculation.

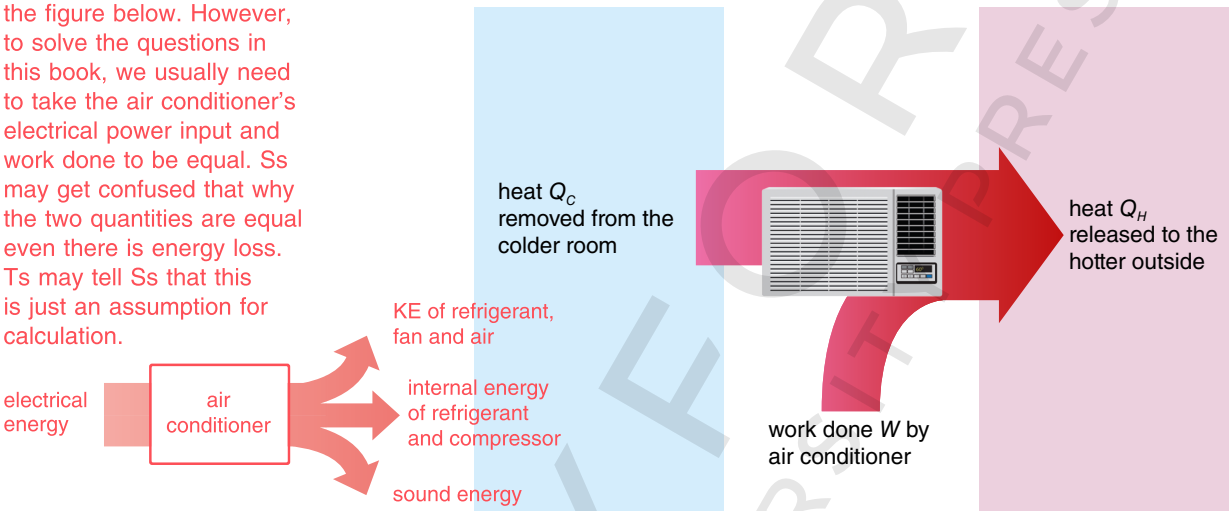


Fig 1.3e $Q_H = Q_C + W$.

Know more

Humidity control by air conditioner

Water vapour condenses more readily when the temperature drops. When an air conditioner is turned on, the evaporator coil is cold. Water vapour in the room condenses on it. This explains why an air conditioner can be used to reduce the humidity of a room.

Know more

Refrigerators

Have you ever thought of cooling the room by opening the refrigerator door? Does it work?

A refrigerator works like an air conditioner. It makes its compartments cold to keep food fresh or drinks cool. To do this, it transfers heat out of the compartments and release heat to the room. Therefore, opening the door of a refrigerator cannot cool the room, but you can feel cooler standing in front of it. Operating a refrigerator would even make the average temperature of the room increase (since $Q_H > Q_C$ as shown in Figure 1.3e).

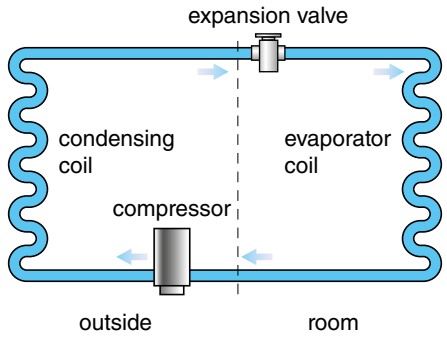


Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

(For Q1–2.) An air conditioner is operating to remove heat from the room to the outside. The figure below shows its schematic diagram.



1 Arrange the following temperatures in ascending order.

p = temperature inside the room

q = temperature outside the room

r = temperature of refrigerant in the evaporator coil

s = temperature of refrigerant in the condensing coil

2 State the temperature changes of the refrigerant when it passes through each of the following components.

(a) Expansion valve

(b) Compressor

Writing practice

Question

Describe the change of state of the refrigerant and the heat exchange when it flows through the evaporator coil in an air conditioner. (2 marks)

Answer

The refrigerant evaporates / changes from liquid to gas. 1A

It absorbs heat from the environment. 1A

Practise yourself

1 Describe the change of state of the refrigerant and the heat exchange when it flows through the condensing coil in an air conditioner. (2 marks)

2 Why is the change of state of the refrigerant important when it flows through the evaporator coil? (2 marks)

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Know more

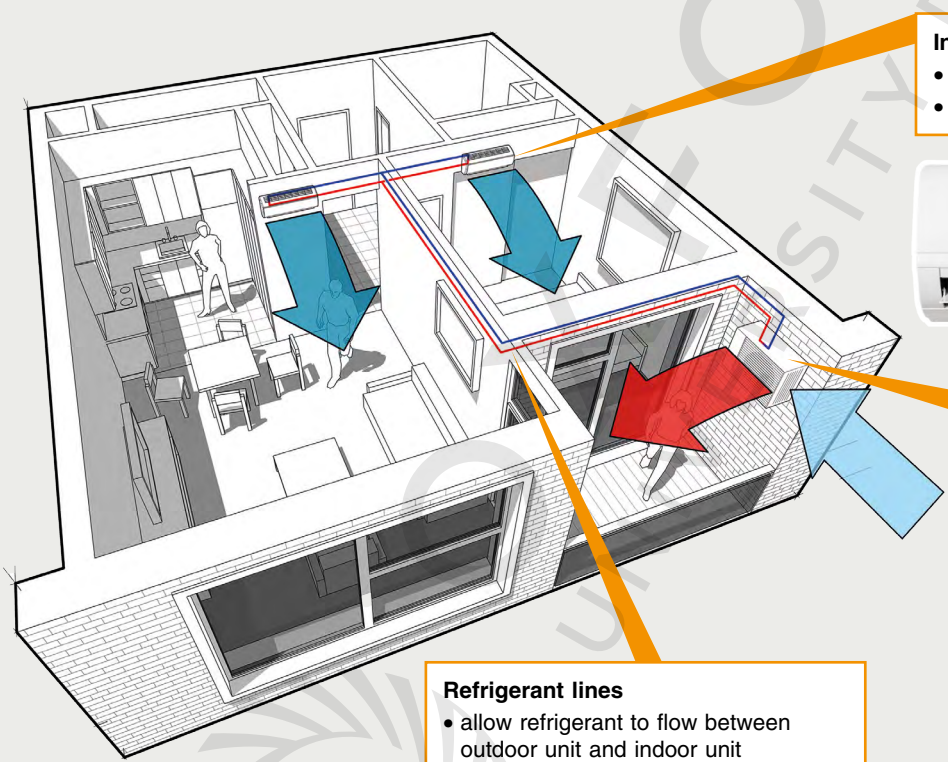
Window-type and split-type air conditioners

In a window-type air conditioner, the evaporator coil and condensing coil are put back-to-back in a single device. This type of air conditioner usually can only serve a room.

In a split-type air conditioner, the evaporator coil and the condensing coil are placed in two separated units. The indoor unit contains the evaporator coil while the outdoor unit contains the compressor (Fig a). An outdoor unit can be connected to several indoor units to cool several rooms (Fig b).



Fig a



Indoor unit
- contains an evaporator coil
- works quietly (without compressor)



Outdoor unit
- contains a condensing coil and a compressor
- can be connected to several indoor units



Refrigerant lines
- allow refrigerant to flow between outdoor unit and indoor unit
- cooling effect may be affected if the lines are too long. In this case, more outdoor units may be needed.

Fig b

Reverse cycle air conditioners (RCAC) is a special type of air conditioner that can either cool or warm air. The warming function can be achieved by reversing the flow of refrigerants. By doing this, energy is absorbed from outside the room and released indoors.

reverse cycle air conditioner 冷暖空調機

☆ HKDSE 2015 Paper 2 Q.3

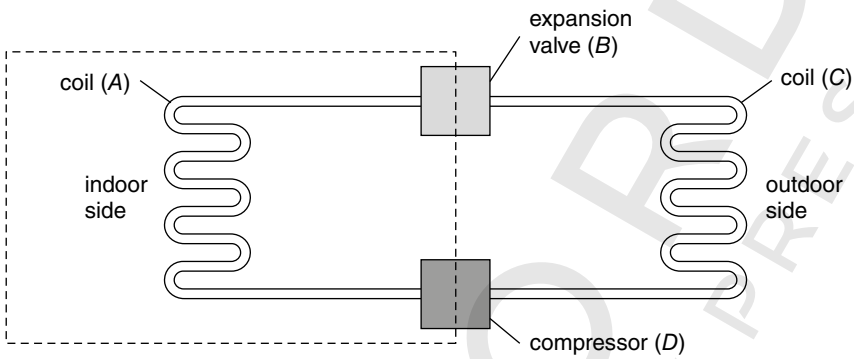
→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Reverse cycle air conditioners

The figure shows the major components of a reverse cycle air conditioner (RCAC), including the expansion valve, the compressor and the coils at indoor and outdoor sides. Refrigerant flows in the loop, passing through the components in cycles. An RCAC can act as a cooler in summer and a heater in winter.

✎ A RCAC is more efficient than a fan heater or oil radiator. It is also safer as it will not cause burns or be knocked over.



Assume the RCAC now acts as a heater.

(a) Describe the functions of the expansion valve and the compressor. (4 marks)

(b) Describe the direction of energy transfer

(i) between the refrigerant and the air at the indoor side,

(ii) between the refrigerant and the air at the outdoor side.

(2 marks)

(c) State the direction of flow of refrigerant, starting from the coil (*A*) at the indoor side. (1 mark)

Solution

(a) The expansion valve allows the refrigerant to enter a region of much lower pressure. 1A

This makes part of the liquid refrigerant evaporate and greatly reduces the temperature of the refrigerant. 1A

The compressor compresses the refrigerant to a higher pressure. 1A

This increases the temperature of the refrigerant. 1A

Common mistake ⚠
Students may not be able to apply their knowledge in air conditioner to RCAC.

(b) (i) Energy is transferred from the refrigerant to the air at the indoor side. 1A

(ii) Energy is transferred from the air at the outdoor side to the refrigerant. 1A

Common mistake ⚠
Students may not understand the working principle of an RCAC and thus make mistakes in the direction of the flow of refrigerant.

(c) $A \rightarrow B \rightarrow C \rightarrow D$ 1A

▶ Revision 1 Q27 (p.42)

DSE goal

Interpret cooling capacity as the rate at which a cooling appliance is capable of removing heat from a room and use kilowatt (kW) as a unit for cooling capacity to solve related problems.

Define coefficient of performance COP as ratio of cooling capacity to electrical power input and solve related problems.

DSE exam

14(2)Q3.2, 15(2)Q.3(a)–(b), 16(2)Q3.7, 18(2)Q3.3, 19(2)Q3.4, 21(2)Q3.6, 22(2)Q3.5, 23(2)Q.3, etc.

✎ Ss should not confuse the energy removed from the room with the energy consumed by the air conditioner.

2 Cooling capacity and coefficient of performance

a Cooling capacity

Cooling capacity is the rate at which an air conditioner is capable of removing heat from a room. The higher the cooling capacity, the faster an air conditioner can remove heat.

$$\text{Cooling capacity} = \frac{\text{maximum amount of heat removed}}{\text{time}} = \frac{Q_C}{t}$$

The common unit of cooling capacity is the kilowatt (kW).

Example 5 Cooling capacity

An air conditioner of cooling capacity 1.2 kW is installed in an empty room of dimensions 3 m × 3 m × 2.5 m. It is turned on for 3 minutes. The specific heat capacity of air is 1000 J kg^{-1} °C^{-1} and the density of air is 1.2 kg m^{-3}.

(a) Find the amount of heat that the air conditioner removes from the room.

(b) Estimate the temperature drop of the air in the room.

(c) The actual temperature drop of the air in the room is smaller than that estimated in (b). Give one reason.

Unless otherwise specified, we assume an air conditioner works at a rate the same as the cooling capacity. ▶

Solution

(a) Amount of heat removed

= cooling capacity × time = 1200 × (3 × 60) = 2.16 × 10^5 J

Tip

Mass = density × volume ▶

(b) Mass of air in the room

= 1.2 × (3 × 3 × 2.5) = 27 kg

By $Q = mc\Delta T$,

$$\Delta T = \frac{Q}{mc} = \frac{2.16 \times 10^5}{27 \times 1000} = 8\ °C$$

The temperature drop is 8 °C.

(c) Heat may be transferred from the outside to the room through walls. / Heat has to be removed from walls.

cooling capacity 冷卻能力 / 製冷能力

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

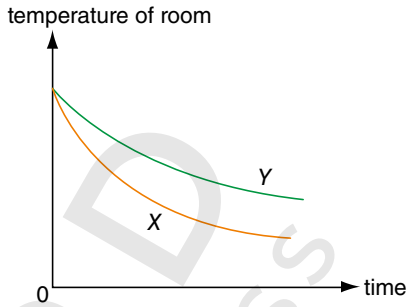
1 The cooling capacity of an air conditioner is

A the maximum amount of heat removed by the air conditioner from indoors per unit time.

B the maximum amount of heat released by the air conditioner to outdoors per unit time.

C the maximum amount of energy consumed by the air conditioner per unit time.

2 There are two identical rooms *X* and *Y* under the same environment. Each room is cooled by an air conditioner. The variation of the temperatures of the rooms with time is as shown.



Which air conditioner has a higher cooling capacity?

A The one in room *X*

B The one in room *Y*

C They have the same cooling capacity

D Cannot be determined

Learn by practice 2

1 An air conditioner has a cooling capacity of 3.8 kW. How long does it take to remove 9.12×10^5 J of heat from a room?

2 An air conditioner removes 0.2 kW h of heat from a room in 5 minutes. Find its cooling capacity.

3 An empty room is cooled by an air conditioner of cooling capacity 2 kW. The mass of air in the room is 40 kg. The specific heat capacity of air is 1000 J kg^{-1} $°C^{-1}$.

(a) Estimate the temperature drop after the air conditioner has been turned on for 3 minutes.

(b) How long does it take for the temperature to drop by 6 °C?

b Coefficient of performance

The energy input of an air conditioner is mainly used in compressing and pumping the refrigerant.

An air conditioner consumes electrical energy to move heat from one place (indoor) to another (outdoor). The **coefficient of performance** (**COP**) is a measure of how efficiently the air conditioner uses energy for moving heat.

$$\text{COP} = \frac{\text{cooling capacity}}{\text{electrical power input}}$$

coefficient of performance (COP) 性能係數

COP
$= \frac{\text{cooling capacity}}{\text{electrical power input}}$
$= \frac{Q \text{ removed per unit time}}{E \text{ input per unit time}}$
$= \frac{Q \text{ removed}}{E \text{ input}}$

The end-use energy efficiency of an air conditioner calculated from its energy output (work done in compressing refrigerant) cannot directly indicate how efficient an air conditioner is in removing heat. Therefore, COP is used to indicate this.
Ss may misinterpret that a COP larger than 1 violates the law of conservation of energy. It can be clarified that the COP is just a ratio of the energy input of air conditioner and the amount of heat removed, and neither of these two quantities comes from part of the other one after some energy conversions.

▶ COP is a ratio and has no unit. The larger the COP, the less energy input is needed to move a certain amount of heat. A typical air conditioner has a COP of 2.3 to 3.2. This means that for each kW h of energy input into the air conditioner, 2.3 to 3.2 kW h of heat can be removed from the room.

Table 1.3b compares the cooling capacity and COP of an air conditioner.

	Cooling capacity	COP
Meaning	the rate of an air conditioner is capable of removing heat from a room	how efficiently an air conditioner uses energy to remove heat
Equation	$\frac{\text{maximum amount of heat removed}}{\text{time}}$	$\frac{\text{cooling capacity}}{\text{electrical power input}}$
Unit	W or kW	no unit

Table 1.3b *Comparing cooling capacity and COP of an air conditioner.*

The following relation can be implied from the law of conservation of energy:

$$\begin{matrix}\text{rate of heat} \\ \text{released to outside}\end{matrix} = \text{cooling capacity} + \begin{matrix}\text{electrical} \\ \text{power input}\end{matrix}$$

Let's see how this relation can be applied to solve problems.

Example 6 | COP

The COP of a 1500-W air conditioner is 2.3.

(a) Find the amount of heat that it can remove from a room in 1 minute.

(b) Find the rate at which heat is being released to the outside.

Solution

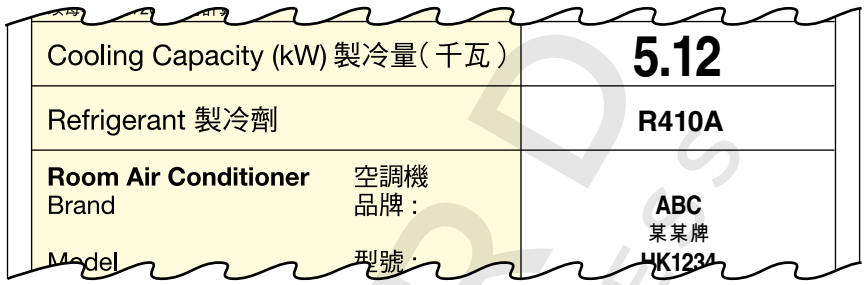
(a) Cooling capacity = electrical power input × COP
$= 1500 \times 2.3$
$= 3450$ W

Amount of heat removed = cooling capacity × time
$= 3450 \times 60$
$= 2.07 \times 10^5$ J

(b) Rate of heat released to outside
= cooling capacity + electrical power input = 3450 + 1500 = 4950 W

Example 7 Energy label of air conditioner

The following is an energy label of an air conditioner.



Cooling Capacity (kW) 製冷量（千瓦）	**5.12**
Refrigerant 製冷劑	**R410A**
Room Air Conditioner 空調機 Brand 品牌：	**ABC** 某某牌
Model 型號：	**HK1234**

If the air conditioner is used for 1200 hours this year and the total energy consumption is 2074 kW h, find the COP of the air conditioner.

Solution

$$P = \frac{E}{t} = \frac{2074}{1200} = 1.73 \text{ kW}$$

$$\text{COP} = \frac{\text{cooling capacity}}{P}$$

$$= \frac{5.12}{1.73}$$

$$= 2.96$$

Quick check 7

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 The cooling capacity of an air conditioner (can / cannot) be larger than its electrical power input.

2 If the coefficient of performance of an air conditioner is 3, it means that

A 1 J of heat is removed from the room for every 3 J of energy consumed by the air conditioner.

B 1 J of energy is consumed by the air conditioner for every 3 J of heat removed from the room.

C 1 J of heat is released to outdoors for every 3 J of energy consumed by the air conditioner.

D 1 J of energy is consumed by the air conditioner for every 3 J of heat released to outdoors.

3 The coefficient of performance of an air conditioner (can / cannot) be larger than 1.

Learn by practice 3

1 An air conditioner with a power input of 540 W releases heat to the outside at a rate of 1890 W. Find its cooling capacity.

2 An air conditioner has a cooling capacity of 2.1 kW. Its input power is 700 W.
(a) Find the COP of the air conditioner.
(b) Find the rate of heat released to the outside.

3 The COP of an air conditioner is 2.4. Find the rate of heat released to the outside when it removes heat from the room at a rate of 1600 W.

4 An air conditioner can remove 2.8 J of heat from a room for every 1 J it consumes. If the air conditioner has consumed 80 kJ of energy, how much heat has been released to the outside?

You may find an online calculator to help you choose an air conditioner with a suitable cooling capacity.

Useful link

▶ To choose a suitable air conditioner for a room, the first thing we consider is usually the size of the room (Table 1.3c). In addition, if a room has a high rate of heat gain from the outside, for example, with a large area of windows or on the highest floor, an air conditioner with a high cooling capacity may be needed.

Room size / square feet	Cooling capacity / BTU h^{-1}
50–70	6500–7200
80–100	8000–9000
110–150	12 000–14 000
160–200	16 000–18 000
210–250	20 000–24 000

Table 1.3c Suggested cooling capacity for different room size (BTU h^{-1} is another unit of cooling capacity, with 1000 BTU h^{-1} = 293 W).

Cooling capacity can also be measured in British thermal units per hour or Btu h^{-1}.

After determining the suitable cooling capacity for a room, we may choose an air conditioner with high COP to save energy.

Other factors, e.g. the refrigerant used, noise level and price, may also be taken into consideration in choosing air conditioners.

Know more

Horsepower

In appliance stores, air conditioners are often rated by their *horsepower*. The horsepower measures only the power input of an air conditioner, not its cooling capacity.

1 horsepower = 746 W

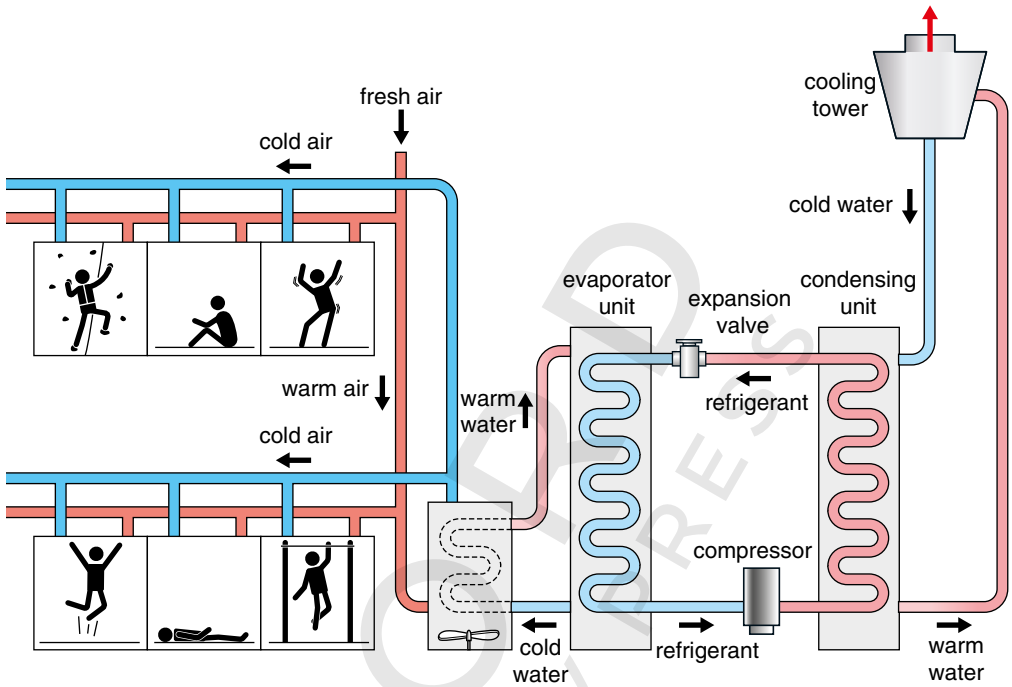
DSE goal

Discuss possible ways of using heat generated by central air conditioning systems.

3 Using heat generated by central air conditioning systems

In many buildings, a central air conditioning system (Fig 1.3f on p.32) is used to cool the whole building and a huge amount of heat removed from the building is ejected into the atmosphere at the cooling tower. If this heat can be used for other purposes, a large amount of energy can be saved.

horsepower 馬力



As water brings away heat more efficiently, less energy is needed to operate the compressor. A water-cooled chiller can save more than 30% of electricity than an air-cooled chiller.

Fig 1.3f *Schematic diagram of a typical water-cooled central air conditioning system. A water-cooled system uses water, instead of air, to carry away heat from the condensing unit. Heat is then ejected out into the atmosphere at the cooling tower as waste heat.*

This system is only applicable if there is a requirement for cooling and heating simultaneously.

In a *heat recovery chiller*, circulating water carries away the heat from the condensing unit and is used to heat space or water for daily usage (Fig 1.3g). Extra heat is then removed from the cooling tower. A heat recovery chiller is suitable for buildings such as hotels, hospitals and government buildings, where stable heating is needed.

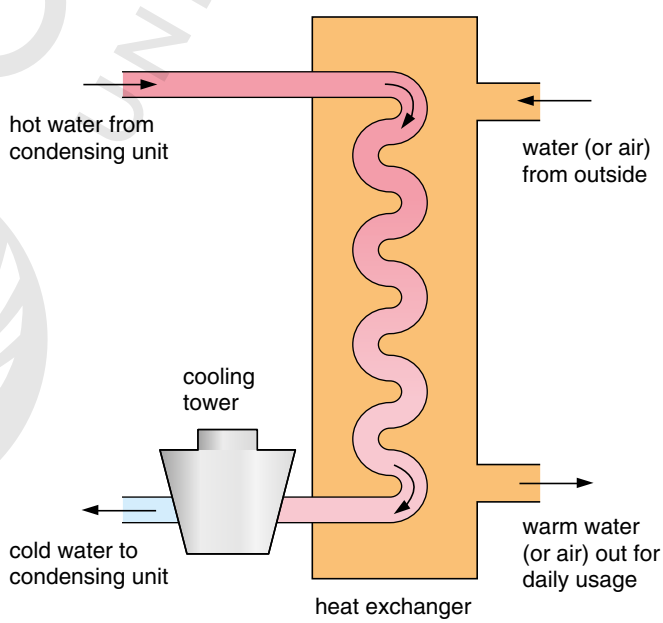


Fig 1.3g *Hot water from the condensing unit is used to heat up water or air for further use.*

Know more

Secret tunnel under HSBC Main Building

There is a very large secret tunnel connected to HSBC Main Building in Central. Do you know its function?

▸ p.44

District cooling system

A large-scale cooling system can even be used to cover a district, e.g. the Kai Tak Development Area.

▸ p.44

heat recovery chiller 熱回收冷水機組

Exercise 1.3

(For Q1–2.) The following table (Table a) shows the input powers and cooling capacities of four air conditioners.

Air conditioner	Input power	Cooling capacity
P	1000 W	3000 W
Q	1200 W	2500 W
R	800 W	2800 W
S	1000 W	2500 W

Table a

1 Which air conditioner is most efficient in using energy to remove heat from a room?

A *P*
B *Q*
C *R*
D *S*

★**2** Which air conditioner releases heat to the outside at the largest rate when operating?

A *P*
B *Q*
C *R*
D *S*

★**3** Which of the following statements about the refrigerant in an air conditioner is/are correct?

(1) It absorbs latent heat of vaporization when it flows through the condensing coil.
(2) Its temperature is higher than that of the room before flowing into the compressor.
(3) It cools down after passing through the expansion valve.

A (1) only
B (3) only
C (1) and (2) only
D (2) and (3) only

★**4** Which of the following statements about the coefficient of performance of an air conditioner is/are correct?

(1) It is a measure of how efficiently an air conditioner uses energy to remove heat.
(2) It will never be greater than 1.
(3) Its value is directly proportional to the input power of the air conditioner.

A (1) only
B (2) only
C (1) and (2) only
D (2) and (3) only

★**5** An air conditioner has a power input of 2 kW and a COP of 2.5. Find the time it needs to remove 4×10^6 J of heat from a room.

★**6** An energy label affixed to an air conditioner is shown below (Fig a).

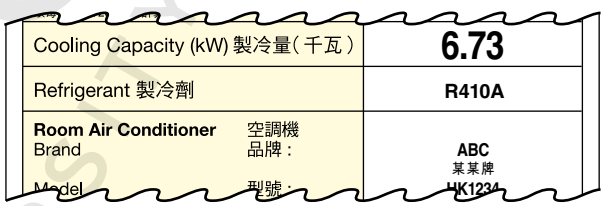


Fig a

The power input of the air conditioner is 910 W. Assume that the air conditioner is used for **720 hours each year**. Estimate

(a) the total energy consumption, and
(b) the total amount of heat removed from a room

in a year. Express the answers in kW h.

★**7** An air conditioner is installed in a room of dimension 4 m × 4 m × 3 m. The power input of the air conditioner is 800 W. After it is turned on, the room temperature drops from 33 °C to 26 °C in 5 minutes. The density of air is 1.2 kg m^{-3} and the specific heat capacity of air is 1000 J kg^{-1} $°C^{-1}$. Assume that the room only contains air and is airtight. Also assume there is no heat transfer between the room and the outside.

(a) Find the mass of air in the room.
(b) Find the amount of heat removed from the room.
(c) Find the COP of the air conditioner.

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*8 Figure b shows the basic components of an air conditioner.

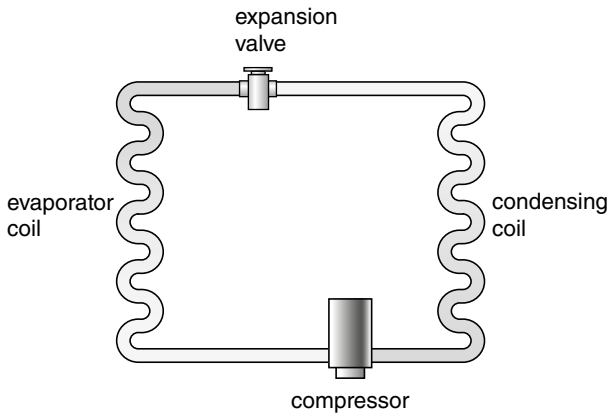


Fig b

(a) Describe the function of the compressor.

(b) Draw arrows in Figure b to show the direction of flow of the refrigerant.

*9 An air conditioner has a cooling capacity of 3.5 kW. It will consume 28 kW h of electrical energy if it is used for 24 hours.

(a) Estimate the power input of the air conditioner.

(b) Estimate the coefficient of performance of the air conditioner.

(c) The air conditioner is used for 9 hours each day. What is the cost of electricity in 30 days? The cost of electricity is $1.1 per kW h.

*10 Refer to the cyclic process in an air conditioner (Fig c).

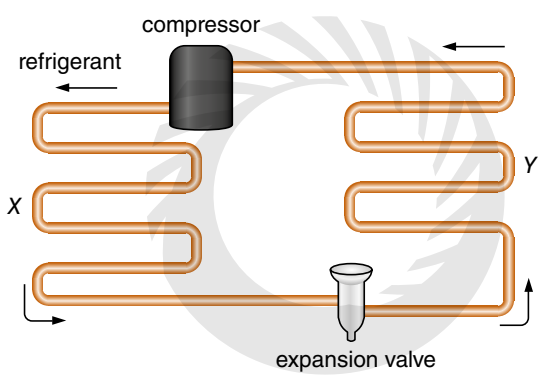


Fig c

(a) Name the parts *X* and *Y*.

(b) Describe how the average kinetic energy and average potential energy of the molecules in the refrigerant change after the refrigerant flows through the expansion valve. Explain your answer.

*11 Figure d shows a schematic diagram of a heat recovery chiller.

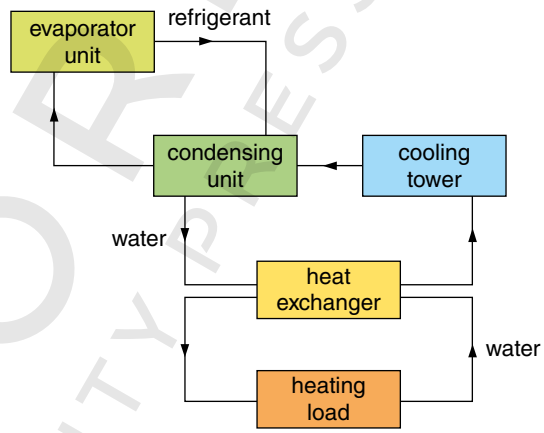


Fig d

(a) State an advantage of using water to carry heat away from the condensing unit.

(b) Describe how the heat recovery chiller reuses the waste heat released by the condensing unit.

(c) Although the waste heat can be reused, a cooling tower is still needed in this system. Explain why.

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Key concepts 1

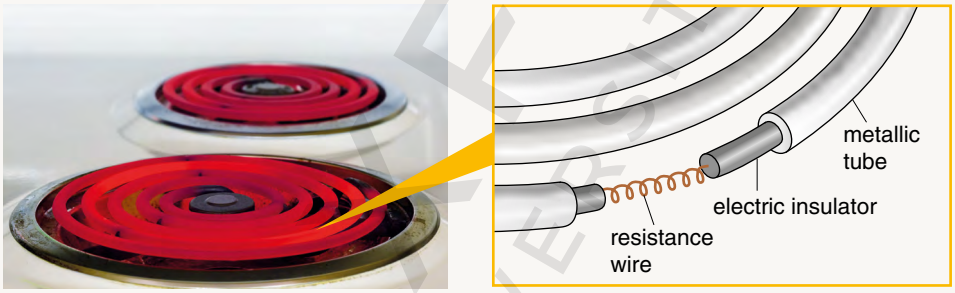
1 Energy efficiency

- end-use energy efficiency = $\frac{\text{useful energy output}}{\text{energy input}} \times 100\%$
- energy label shows the energy performance of an appliance

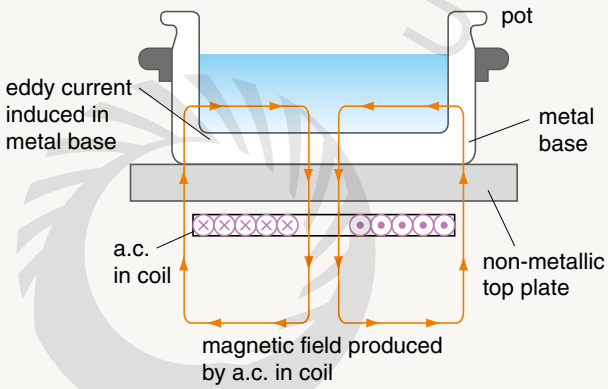
2 Cooking without fire

Electric hotplate

- heating effect of current through resistance wire



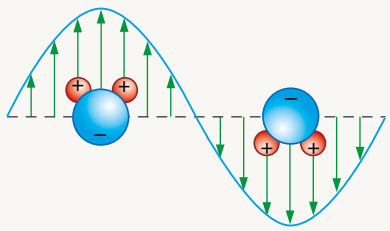
Induction cooker



- varying *B*-field produced by a.c. in coil

  ⇒ eddy currents induced in metal pot produce heating effect

Microwave oven

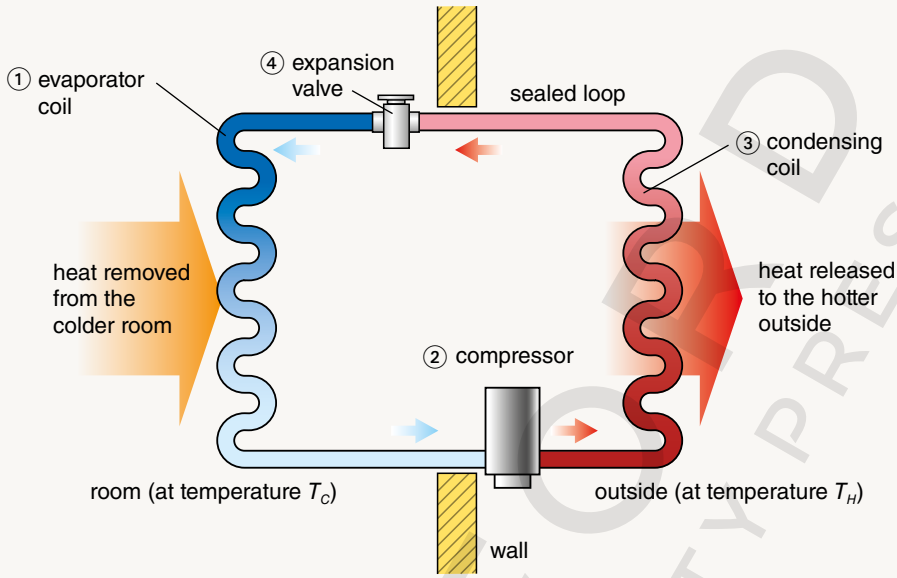


- polar water molecules flip and gain energy as microwaves pass through

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3 Air conditioner

- heat pump transfers heat against its natural direction of flow



① cool refrigerant absorbs heat from air and evaporates
⇒ cooling the room

② refrigerant is compressed and becomes much hotter than outside

③ hot refrigerant releases heat to outside and condenses

④ refrigerant at low pressure, partly evaporates and cools down

Cooling capacity

- rate of an air conditioner of removing heat from a room

$$\text{cooling capacity} = \frac{Q_C}{t}$$

- unit: kW

Coefficient of performance (COP)

- a measure of how efficiently the air conditioner uses energy for removing heat

$$\text{COP} = \frac{\text{cooling capacity}}{\text{electrical power input}}$$

- no unit

Key terms

1	coefficient of performance (COP) 性能係數	p.28
2	cooling capacity 冷卻能力 / 製冷能力	p.27
3	electric hotplate 電熱平板爐	p.9
4	end-use energy efficiency 最終能源效益	p.3
5	Energy Efficiency Labelling Scheme (EELS) 能源效益標籤計劃	p.5
6	heat pump 抽熱機	p.20
7	induction cooker 電磁爐	p.9
8	microwave oven 微波爐	p.9

Revision 1

Multiple-choice questions

★1 1.1 The energy conversions of an electric fan are shown below (Fig a).

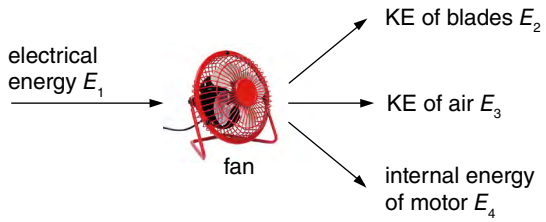


Fig a

What is the end-use energy efficiency of the fan?

A $\frac{E_3}{E_1} \times 100\%$

B $\frac{E_2 + E_3}{E_1} \times 100\%$

C $\frac{E_3}{E_1 + E_2} \times 100\%$

D $\frac{E_1 - E_3}{E_1} \times 100\%$

★2 1.3 Which of the following statements about the cooling capacity of an air conditioner is/are correct?

(1) It is the rate at which heat is released to the outside by the air conditioner.

(2) Its unit is kW h.

(3) Its value can be larger than the power input of the air conditioner.

A (1) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

★3 1.3 Which of the following air conditioners releases heat to the outside at the largest rate?

	Air conditioner	Input power	COP
A	*P*	0.8 kW	2
B	*Q*	0.9 kW	1.6
C	*R*	1.0 kW	1.5
D	*S*	1.1 kW	1.2

★4 1.3 What happens to the refrigerant when it flows through the compressor in an air conditioner?

(1) Its volume decreases.

(2) Its pressure increases.

(3) Its temperature increases.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★5 1.3 What happens to the refrigerant when it flows through the evaporator coil in an air conditioner?

(1) Its internal energy increases.

(2) Its volume increases.

(3) It absorbs energy from the room.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★6 1.2 Which of the following descriptions about different electric cookers is/are correct?

(1) In using an electric hotplate, heat is transferred from the coil of resistance wire to the cooking pot.

(2) In using an induction cooker, heat is transferred from the cooking pot to the top plate.

(3) In using a microwave oven, heat is transferred from the inner case of the oven to the food.

A (1) only

B (1) and (2) only

C (1) and (3) only

D (2) and (3) only

★7 1.3 An air conditioner operating at power input *P* has a coefficient of performance (COP) of 3. What amount of heat does it release to the outside in time *t*?

A *Pt*

B Between *Pt* and 3*Pt*

C 3*Pt*

D Greater than 3*Pt*

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★8 1.1 The following energy label is affixed to a television (Fig b).

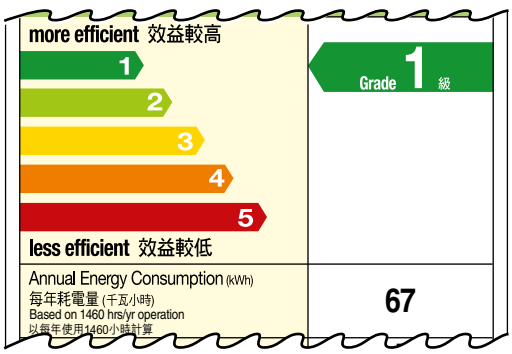


Fig b

Which of the following can be concluded from the label?

(1) The power of the television is about 46 W.

(2) The energy performance of the television is below average in its category.

(3) The standby power of the television is 1 kW.

A (1) only B (2) only

C (1) and (2) only D (1), (2) and (3)

★9 1.1 Two electric kettles have the same power output. They are used to boil the same amount of water from the same temperature. Which of the following statements is **incorrect**?

A The one with lower power input consumes less electricity.

B The one with higher end-use energy efficiency boils the water faster.

C The one with lower end-use energy efficiency consumes more electricity.

D The one with lower end-use energy efficiency loses more heat to the surrounding.

★10 1.1 Appliances are graded in EELS by calculating their energy efficiency index I_ε.

$$I_\varepsilon = \frac{E_{ac}}{E_{av}} \times 100\%$$

where E_{ac} is the actual energy consumption of the product, and E_{av} is the average energy consumption of the products in the same category.

The grading scheme for electric storage water heaters is shown in Table a.

I_ε (%)	Grade
$I_\varepsilon \leq 75$	1
$75 < I_\varepsilon \leq 90$	2
$90 < I_\varepsilon \leq 105$	3
$105 < I_\varepsilon \leq 120$	4
$120 < I_\varepsilon$	5

Table a

At least by how many percent is the annual energy consumption of a water heater rated as grade 2 lower than that rated as grade 4?

A $\frac{105 - 75}{105} \times 100\% = 28.6\%$

B $\frac{105 - 90}{105} \times 100\% = 14.3\%$

C $\frac{120 - 75}{120} \times 100\% = 37.5\%$

D $\frac{120 - 90}{120} \times 100\% = 25\%$

★★11 1.3 An air conditioner of cooling capacity 2.7 kW is used to cool a room. The graph below shows how the temperature T of the room varies with time t (Fig c).

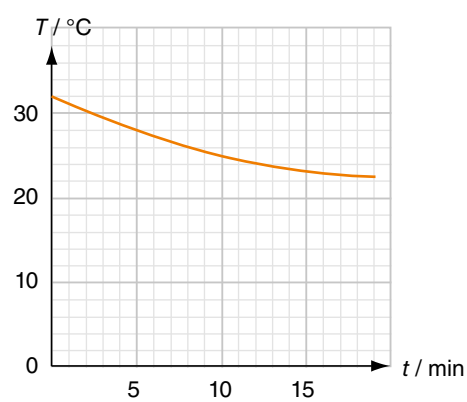


Fig c

The coefficient of performance (COP) of the air conditioner is 3. The cost of electricity is \$1.1 per kW h. What is the cost of the electricity in cooling the room from 32 °C to 24 °C?

A \$0.083

B \$0.198

C \$0.594

D \$0.713

Refer p.28–29

12 **HKDSE 2015 Paper 2 Q3.4**
1.1

A coal-fired power plant generates and transmits electrical energy to consumers a long distance away. When a consumer connects an incandescent lamp to the mains supply, the following data illustrates what happens per 1000 J of energy in coal converted to electrical energy which is then supplied to the lamp.

Loss during generation process in the power plant	600 J
Loss during transmission before reaching the lamp	100 J
Heat generated by the lamp	250 J
Visible light generated by the lamp	50 J

What is the **end-use energy efficiency** of this incandescent lamp?

A 5% B 12.5%
C 16.7% (21%) D 30%

13 **HKDSE 2018 Paper 2 Q3.3**
1.3

An air conditioner with $\frac{\text{cooling capacity}}{\text{electrical power input}} = 2$ has a cooling capacity of 746 W. Estimate the rate at which thermal energy is being released to the environment outside.

A 373 W B 746 W
C 1119 W (37%) D 1492 W

14 **HKDSE 2018 Paper 2 Q3.4**
1.2

Which of the following contribute(s) to cooking with a microwave oven?

(1) The energy of microwave is only absorbed by the food's surface and then transferred to its interior by conduction.

(2) Water molecules are polar due to non-uniform charge distribution within each molecule.

(3) The electric field of microwave is continuously changing.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only (37%)

Conventional questions

★**15** Vivian uses an induction cooker to have a hotpot dinner (Fig d).
Misc



Fig d

(a) Describe how the electrical energy supplied to the induction cooker becomes the internal energy of the food. (3 marks)

(b) The top plate of an induction cooker gets hot after using it for a hotpot dinner. Explain why. (1 mark)

(c) In the hotpot dinner, the induction cooker operates at 1200 W in average for 2 hours. The cost of electricity is $1.1 per kW h. Find the cost of electricity for the cooker. (2 marks)

★**16** A bowl of soup is heated by a microwave oven.
1.2

(a) Describe how the soup is heated up by microwaves. (2 marks)

(b) A conventional oven emits infra-red radiation to heat up food. Explain why the middle of the soup gets hot more quickly when using a microwave oven than a conventional oven. (1 mark)

(c) The container should not be closed tightly when reheating the soup using a microwave oven (Fig e). Explain. (2 marks)



Fig e

★17 1.1 The energy label affixed to a household storage water heater is shown in Figure f.

Annual Standby Loss Energy Consumption (kWh) 每年耗電量 (千瓦小時) Based on idling time of 1800 hrs/yr 以每年備用時間為1800小時計算	48
Storage Capacity (litre) 儲水容量（公升）	25
Storage Type Electric Water Heater 儲水式 電熱水器	
Brand 品牌：	ABC 某某牌
Model 型號：	HK1234
Reference Number / Year 參考編號 / 年份：	E200123 / 2020
Information Provider 資料提供者：	XYZ 某某某

Fig f

(a) What is the average power consumed by the heater in standby mode? (2 marks)

(b) What is the amount of energy needed to heat 25 L of water from 15 °C to 65 °C? The density of water is 1 kg L^{-1} and the specific heat capacity of water is 4200 J kg^{-1} $°C^{-1}$. (2 marks)

(c) What is the average power output of the heater in heating the water? The time required to heat the water from 15 °C to 65 °C is 50 minutes. (1 mark)

★18 Misc A 1200-W microwave oven and a 2000-W induction cooker are used to heat water. Their end-use energy efficiencies are 55% and 85% respectively.

(a) Give a reason to explain why a microwave oven has a lower end-use energy efficiency than an induction cooker. (1 mark)

(b) Estimate

(i) the time needed, and

(ii) the cost of electricity

for each cooker to heat 1.5 kg of water from 20 °C to 80 °C. Electricity costs $1.1 per kW h. (5 marks)

★19 Misc An electric hotplate is used to heat a pot of soup (Fig g).



Fig g

(a) State one design of the electric hotplate that

(i) facilitates heat transfer, (1 mark)

(ii) avoids current flowing from the resistance wire to the pot. (1 mark)

(b) The soup has an initial temperature of 20 °C and a heat capacity of 7000 J $°C^{-1}$. During heating, a current of 9 A passes through the resistance wire of resistance 25 Ω for 5 minutes. The end-use energy efficiency of the hotplate is 75%.

(i) Find the energy input to the hotplate. (2 marks)

(ii) Find the final temperature of the soup. (2 marks)

(c) Suggest a reason why an electric hotplate usually has a lower end-use energy efficiency than an induction cooker. (2 marks)

★20 1.1 The following labels are affixed on air conditioners *X* and *Y* (Fig h).

5 less efficient 效益較低	
Annual Energy Consumption (kWh)(Cooling) 每年耗電量 (千瓦小時)(製冷) Based on 1200 hrs/yr operation 以每年使用1200小時計算	1056
Cooling Capacity (kW) 製冷量（千瓦）	6.4

Air conditioner X, Price: $3400

5 less efficient 效益較低	
Annual Energy Consumption (kWh)(Cooling) 每年耗電量 (千瓦小時)(製冷) Based on 1200 hrs/yr operation 以每年使用1200小時計算	596
Cooling Capacity (kW) 製冷量（千瓦）	6.4

Air conditioner Y, Price: $4000

Fig h

Assume that the air conditioners are used for 840 hours each year.

(a) Find the energy consumption of each air conditioner in a year. (3 marks)

(b) If air conditioner *Y* is bought, how many years will it take to pay back the difference in price between the two air conditioners with the savings on electricity? The cost of electricity is $1.1 per kW h. (3 marks)

★21 1.3 An air conditioner with a cooling capacity of 3 kW is switched on to cool a room.

(a) When the liquid refrigerant passes through the expansion valve, part of it evaporates. How does the temperature of the refrigerant change? Why is this process important? (2 marks)

(b) Then the refrigerant flows through the evaporator coil and the remaining liquid refrigerant evaporates.

(i) Describe the heat exchange during this process. (1 mark)

(ii) Estimate the mass of refrigerant that evaporates in each second. Assume the temperature of the refrigerant remains unchanged during this process. The refrigerant used has a specific latent heat of vaporization of 220 kJ kg^{-1}. (2 marks)

(c) Heat is released to the outside by the air conditioner at a rate higher than its cooling capacity. Explain briefly why this is the case. (2 marks)

★22 1.1 The following energy labels are affixed to two washing machines *X* and *Y* (Fig i).

Annual Energy Consumption (kWh) (Washing) 每年耗電量（千瓦小時)(洗滌) Based on 260 washes/yr operation 以每年使用260次計算	299
Washing Capacity (kg) 洗衣量（公斤）	7
Water Consumption (litre) 耗水量（公升）	52

Washing machine X, Price: $3920

Annual Energy Consumption (kWh) (Washing) 每年耗電量（千瓦小時)(洗滌) Based on 260 washes/yr operation 以每年使用260次計算	236
Washing Capacity (kg) 洗衣量（公斤）	7
Water Consumption (litre) 耗水量（公升）	50

Washing machine Y, Price: $4200

Fig i

(a) Describe the energy conversion in operating a washing machine. (2 marks)

(b) Ron uses the washing machine 170 times a year on average.

(i) Estimate the energy (in kW h) consumed by each washing machine in one wash. (2 marks)

(ii) Estimate the amount of money that Ron would save if he used the washing machine *Y* instead of *X* for 10 years. Remember to include the price difference of the washing machines. The cost of electricity is $1.1 per kW h. (2 marks)

★23 Misc An air conditioner of power input 800 W is switched on to cool a living room (Fig j).



Fig j

The temperature of the room drops from 33 °C to 25 °C in 10 minutes. The dimensions of the room are 5 m × 6 m × 3 m. The density of air is 1.2 kg m^{-3}. The specific heat capacity of air is 1040 J kg^{-1} $°C^{-1}$. Assume that the room only contains air and is airtight. Also assume that the heat transfer between the room and the outside is negligible.

(a) Explain why an air conditioner is usually installed high on a wall. (2 marks)

(b) Estimate the mass of air in the living room. (1 mark)

(c) (i) Estimate the cooling capacity of the air conditioner. (2 marks)

(ii) Estimate the coefficient of performance of the air conditioner. (2 marks)

(d) If the heat transfer between the room and the outside is **not** negligible, how will the actual cooling capacity of the air conditioner be different from the answer in (c)(i)? Explain your answer. (2 mark)

(e) The cost of electricity is $1.1 per kW h. Estimate the cost of electricity of the above cooling process. (1 mark)

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*24 1.3 Figure k shows the cyclic process in an air conditioner.

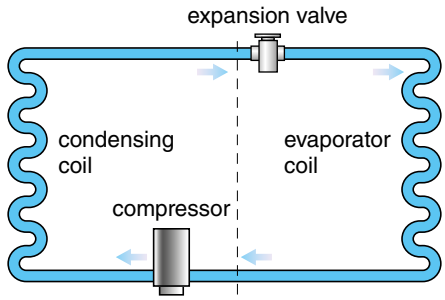


Fig k

(a) Part of the refrigerant evaporates after passing through the expansion valve. Explain, in terms of molecular motion, how the temperature of the liquid refrigerant changes in this process. (3 marks)

(b) State and explain a design of the evaporator coil to facilitate heat transfer. (2 marks)

(c) On a hot summer day, an air conditioner is switched on for some time to cool the room. The energy label of the air conditioner is shown below (Fig l).

Cooling Capacity (kW) 製冷量（千瓦）	**1.98**
Refrigerant 製冷劑	**R410A**
Room Air Conditioner 空調機 Brand 品牌：	**ABC** 某某牌
Model 型號：	**HK1234**

Fig l

The room is cooled from 33 °C to 25 °C and 454 kJ of energy is removed from it.

(i) Estimate the time needed. (1 mark)

(ii) The power input of the air conditioner is 570 W. Estimate the cost of electricity. Electricity costs $1.1 per kW h. (2 marks)

*25 Misc This question is about electric hotplate and induction cooker.

(a) Both cookers make use of the heating effect of electric current for cooking. Where does the current flow to produce the heating effect in each case? (2 marks)

(b) Hence explain why an induction cooker usually has a higher end-use energy efficiency than an electric hotplate. (2 marks)

(c) Both cookers contain a coil of wire. Can each cooker be operated by passing a steady d.c. through the coil? Explain briefly. (4 marks)

(d) An electric hotplate and an induction cooker have end-use energy efficiencies of 70% and 80% respectively. What is the difference in cost in using them to provide energy of 1 kW h for cooking? The cost of electricity is $1.1 per kW h. (2 marks)

**26 1.3 An air conditioner is used to cool a room.

(a) What is the meaning of the cooling capacity of an air conditioner? (1 mark)

(b) Explain how the room is cooled when refrigerant flows through the evaporator coil. (3 marks)

(c) The cost of electricity of using the air conditioner for a month is $264. The cost of electricity per kW h is $1.1. The coefficient of performance (COP) of the air conditioner is 2.5. Estimate the total amount of heat (in kW h) that has been removed from the room in this month. (3 marks)

(d) One day, the compressor fails to function properly. The refrigerant is not compressed before flowing into the condensing coil. Explain what will happen. [Hint: Consider the temperature difference between the refrigerant and the outside.] (3 marks)

Refer p.21–22

27 Misc **HKDSE 2015 Paper 2 Q.3**

Figure m shows the Energy Efficiency Label of an air conditioner *X*.

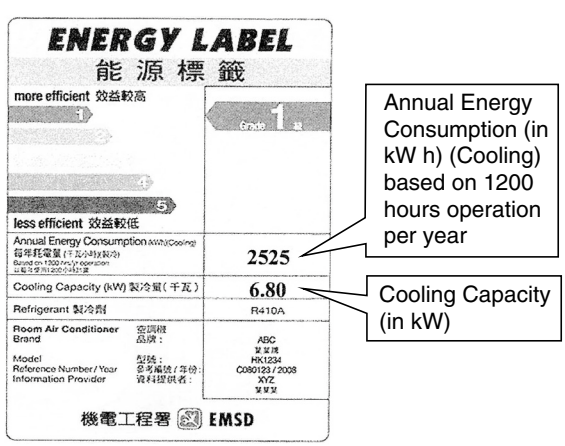


Fig m

(a) This air conditioner is installed in a room of floor area of 20.0 m^2 and height of 3.0 m. Given: density of air = 1.2 kg m^{-3}, specific heat capacity of air = 1000 J kg^{-1} $°C^{-1}$

(i) Estimate the time required to cool the room from 33 °C to 25 °C. Assume that the density and specific heat capacity of air remain constant throughout. (2 marks)

(ii) Suggest a reason why the actual time for cooling from 33 °C to 25 °C is longer than the result found in (a)(i). (1 mark)

(b) (i) Find the average electrical power input (in kW) of air conditioner *X* during operation. (1 mark)

(ii) Find the value of $\frac{\text{cooling capacity}}{\text{electrical power input}}$ for this air conditioner. A student comments that this ratio having a value greater than 1 violates the principle of conservation of energy because the amount of heat removed by the air conditioner is greater than the electrical power input. Discuss the student's comment. (3 marks)

Q27 Exam Report: Part (a)(i) was well answered. (a)(ii) revealed that many candidates were weak in explaining principles of physics. Quite a number of them stated that longer cooling time was due to 'heat lost to surroundings' but not the fact that thermal energy is being removed. In (b), most were able to work out the average electrical power input and the COP. However, many did not fully understand the concept that 1 J of electrical energy consumed by the air-conditioner/compressor can remove 3.24 J of thermal energy from the room. In (c)(i), quite a number of the candidates made mistakes in the direction of the flow of refrigerant. This revealed that they were not familiar with the working principle of a 'reverse-cycle air-conditioner' (RCAC) and the functions of different components in it. As a result, many failed to state in which component the refrigerant had the highest temperature. Part (c)(ii) was generally well answered despite some wrong statements such as reversing the direction of electric current could convert a cool-only air-conditioner into an RCAC.

(c) *X* is a cool-only air conditioner as it can only cool air. Nowadays, 'reverse-cycle air conditioners' (RCAC) that can either cool or warm air are available in the market and they have the same major components of *X*. Figure n shows a simplified schematic diagram of an RCAC with four components *A*, *B*, *C* and *D*, in which *A* is the expansion valve and *C* is the compressor.

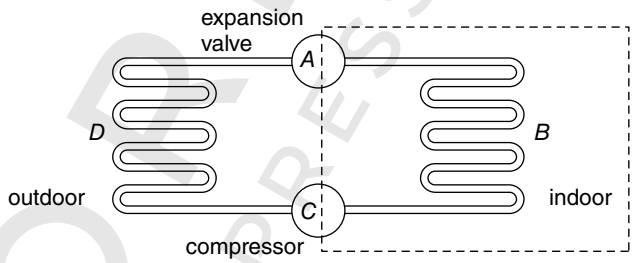


Fig n

(i) If the RCAC operates in a mode **to warm air** during winter time, give the direction of the flow of refrigerant starting from compressor *C* using the letters *A*, *B*, *C* and *D*. State in which component, *A*, *B* or *D*, the refrigerant has the highest temperature. (2 marks)

(ii) Suggest **one** modification that can convert a cool-only air conditioner into an RCAC. (1 mark)

Know more

Secret tunnel under HSBC Main Building (p.32)

There is a very large secret tunnel connected to HSBC Main Building in Central. It contains water pipes which bring seawater in to cool down the central air conditioning system in the building. Up to 1000 L per second of water can flow through the tunnel—enough to fill 34 Olympic-size swimming pools in one day!



Water pipes in the tunnel.



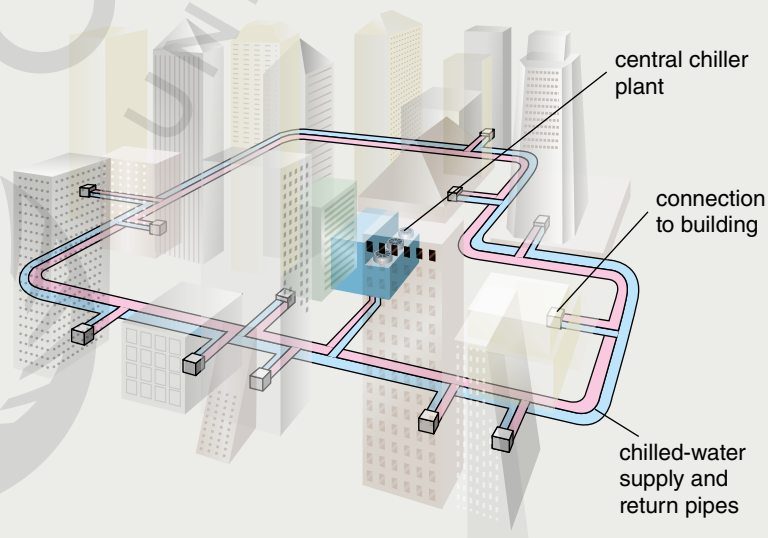
HSBC Main Building.

District cooling system (p.32)

In a district cooling system, a central chiller plant supplies chilled water to the buildings in a district via an underground water-pipe network. The chilled water carries heat away from each building and returns to the central plant.

Such a large-scale system can reduce the total power used in air conditioning and help reduce environmental pollution. Users do not need to install individual chillers. They only need to pay the fee for the chilled water.

The first district cooling system in Hong Kong is located at the Kai Tak Development Area. It provides a cooling capacity up to 462 MW to serve more than 50 buildings. By 2020, it is estimated that the system had saved 20 million kW h of electricity .



A district cooling system.